Assignment 5: Creating a VPC with Subnets

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Deliverables

Each student must submit the following:

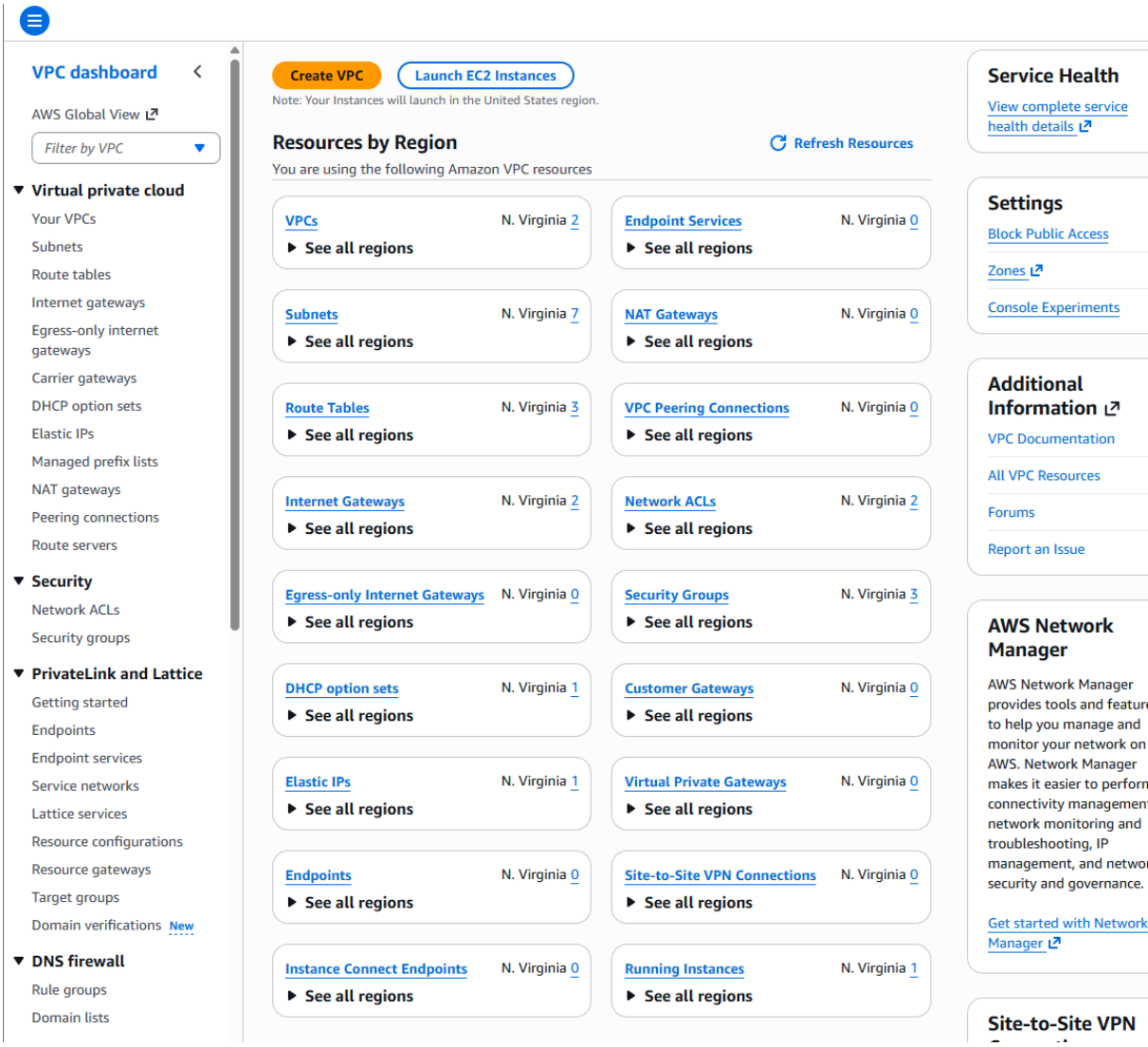
1. Presentation Slides (15–30 slides maximum) ○ Title slide with

   student name(s), course, and assignment title.

   - Clear explanation of each completed task: (Access AWS academy: Module 5 Networking and Content Delivery - Lab 2: Build your VPC and Launch a Web Server)

1. Task 1- Create a VPC.

In the search box to the right of **Services**, search for and choose **VPC** to open the VPC console.

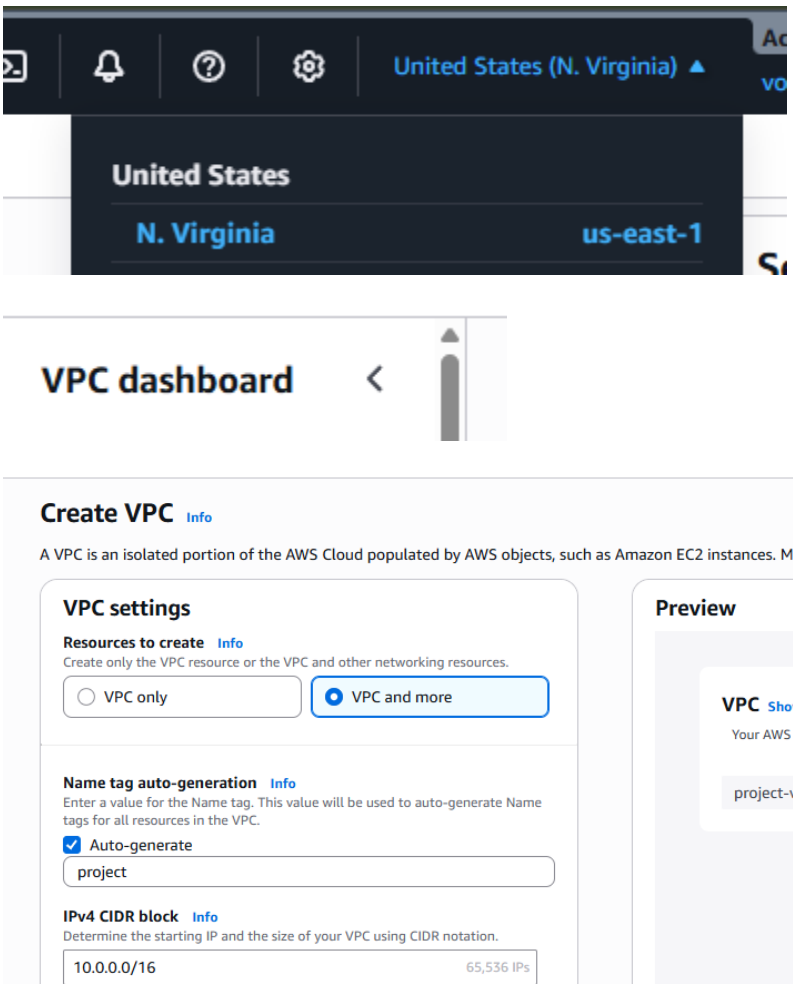


Begin creating a VPC.

In the top right of the screen, verify that N. Virginia (us-east-1) is the region.

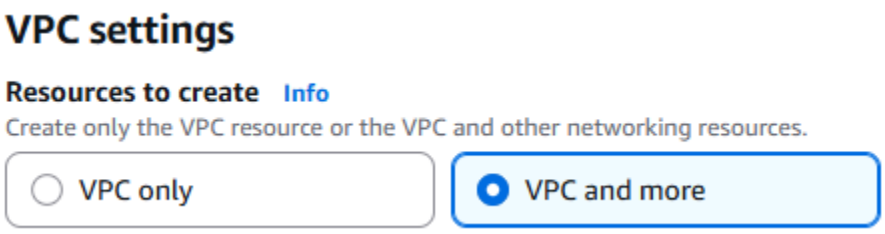
Choose the VPC dashboard link which is towards the top left of the console.

Next, choose Create VPC.

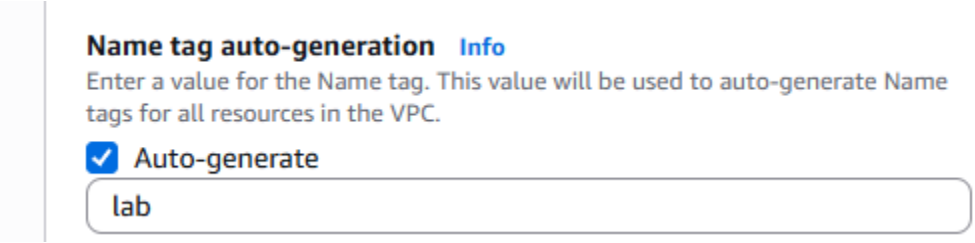


Configure the VPC details in the VPC settings panel on the left:

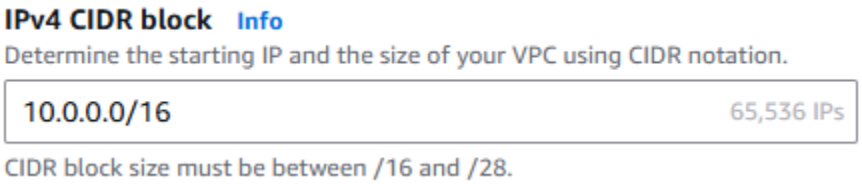
Choose VPC and more.



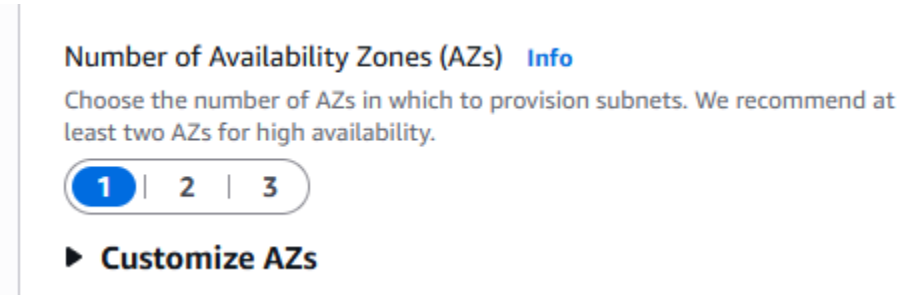
Under Name tag auto-generation, keep Auto-generate selected, however change the value from project to lab.



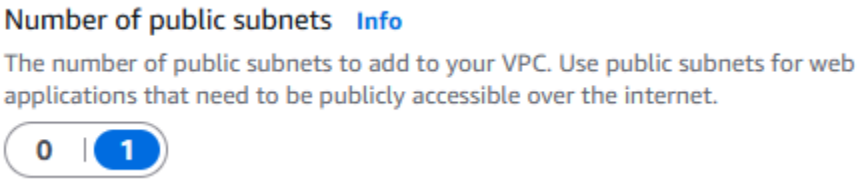
Keep the IPv4 CIDR block set to 10.0.0.0/16



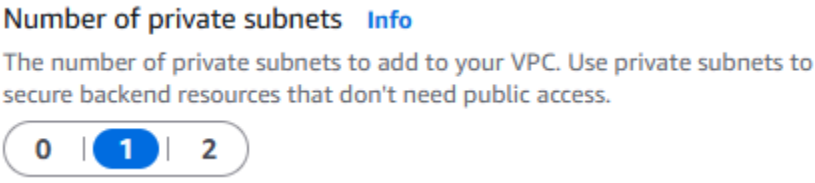
For Number of Availability Zones, choose 1.



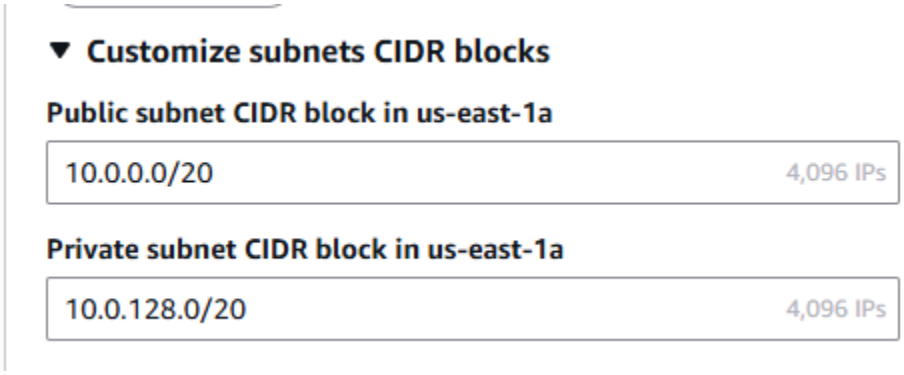
For Number of public subnets, keep the 1 setting.



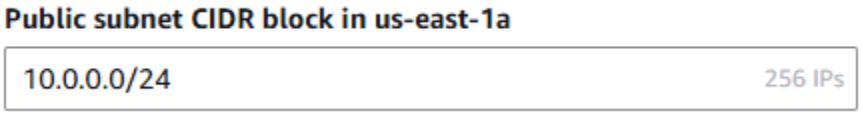
For Number of private subnets, keep the 1 setting.



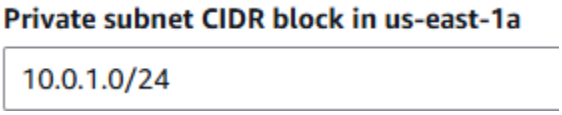
Expand the Customize subnets CIDR blocks section



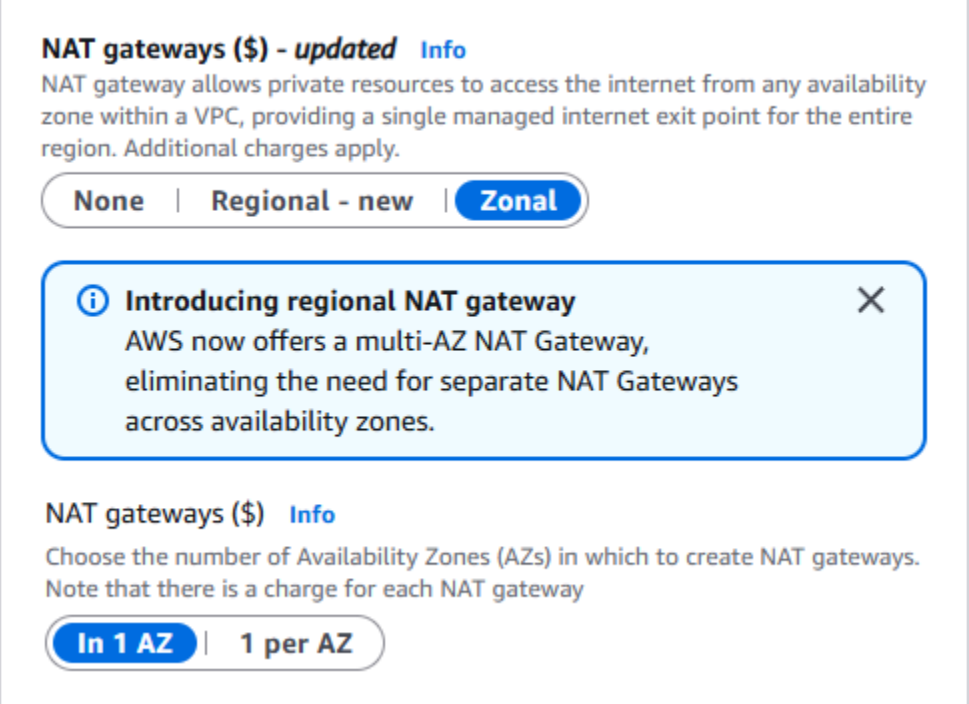
Change Public subnet CIDR block in us-east-1a to 10.0.0.0/24



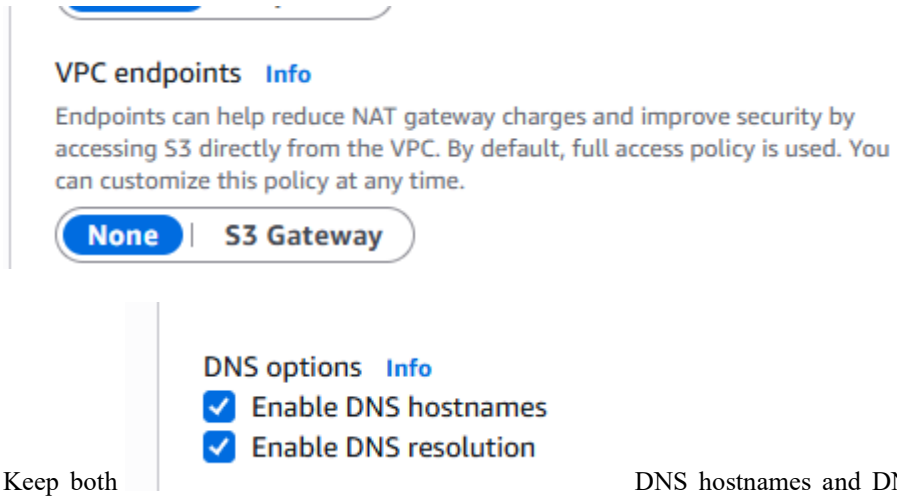
Change Private subnet CIDR block in us-east-1a to 10.0.1.0/24



Set NAT gateways to In 1 AZ.



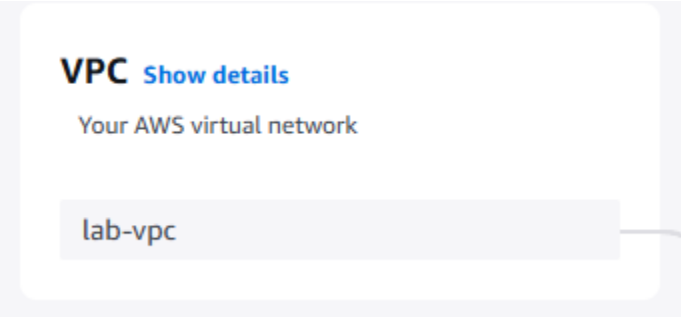
Set VPC endpoints to None.



Keep both DNS hostnames and DNS resolution enabled.

In the Preview panel on the right, confirm the settings you have configured.

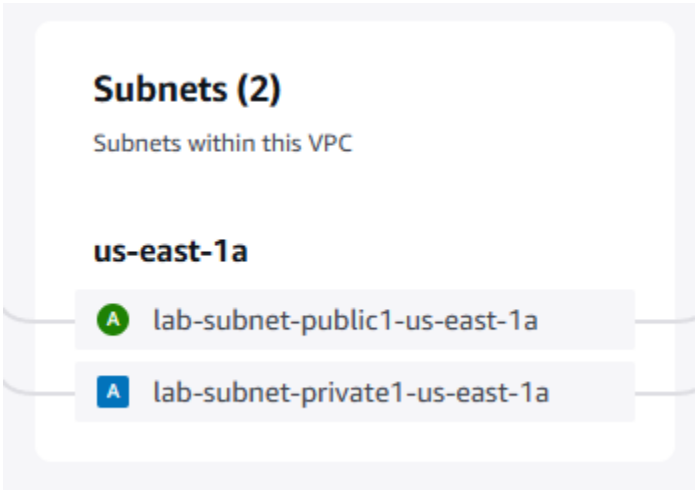
VPC: lab-vpc



Subnets:us-east-1a

Public subnet name: lab-subnet-public1-us-east-1a

Private subnet name: lab-subnet-private1-us-east-1a



Route tables

lab-rtb-public

lab-rtb-private1-us-east-1a



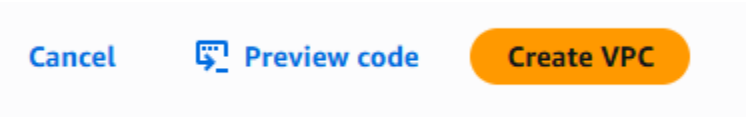
Network connections

lab-igw

lab-nat-public1-us-east-1a

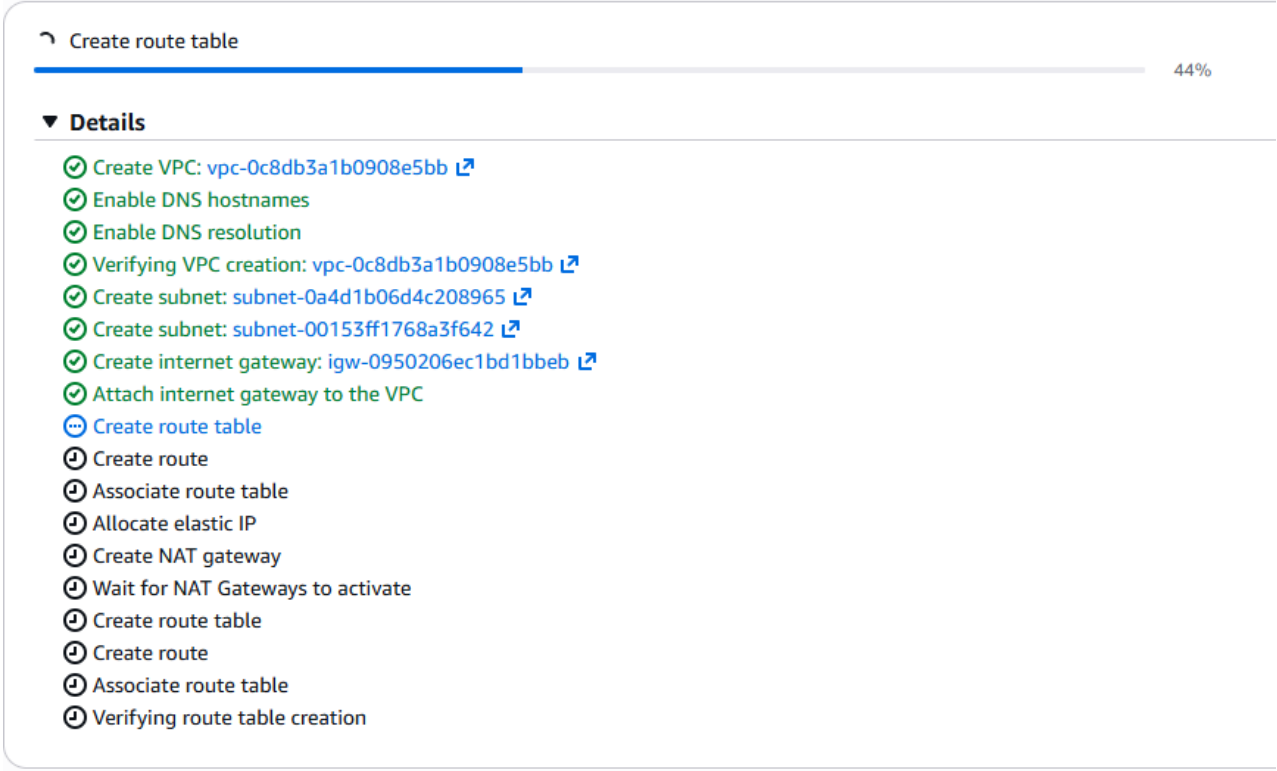


At the bottom of the screen, choose Create VPC



The VPC resources are created. The NAT Gateway will take a few minutes to activate.

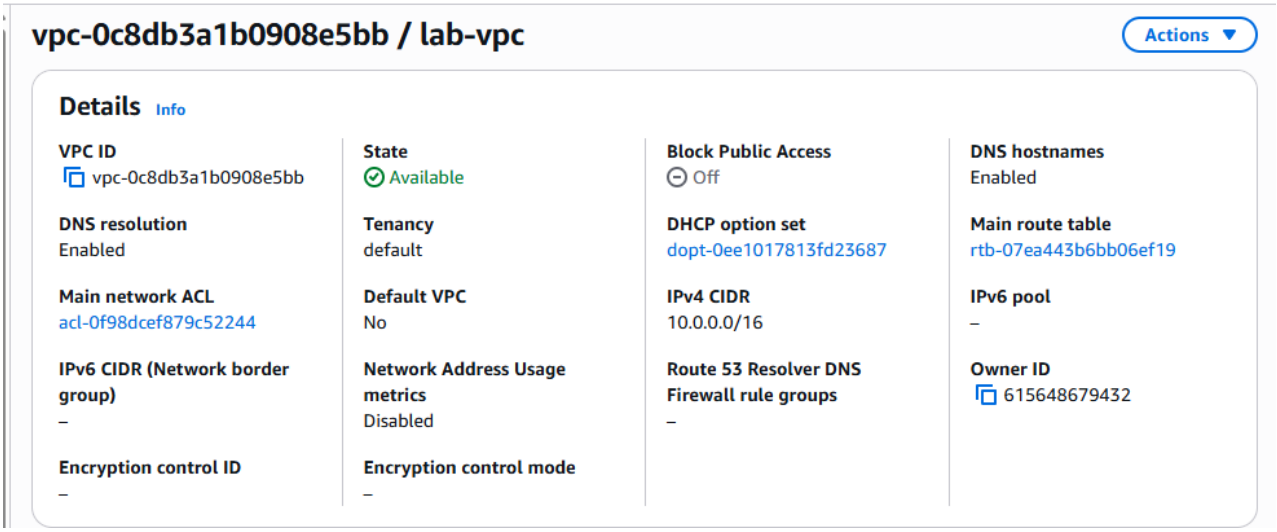
Please wait until all the resources are created before proceding to the next step

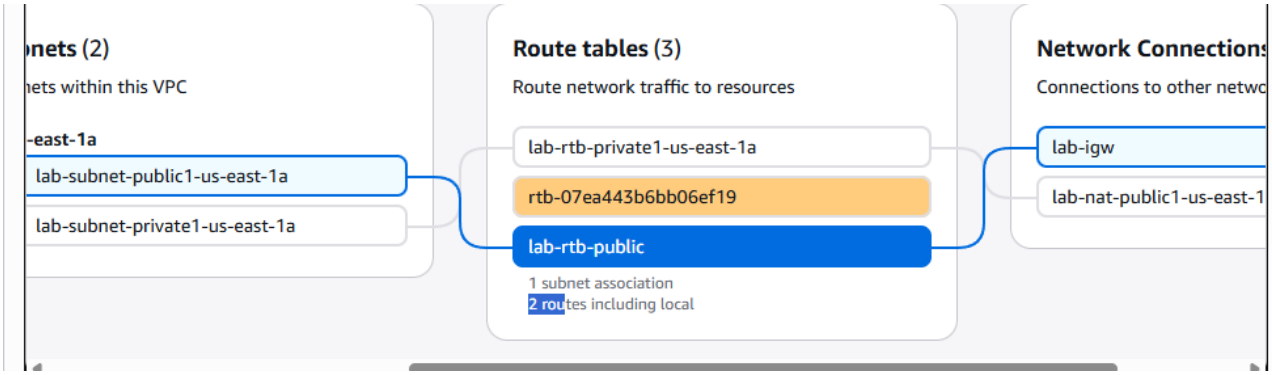


Once it is complete, choose View VPC

The wizard has provisioned a VPC with a public subnet and a private subnet in one Availability Zone with route tables for each subnet. It also created an Internet Gateway and a NAT Gateway.

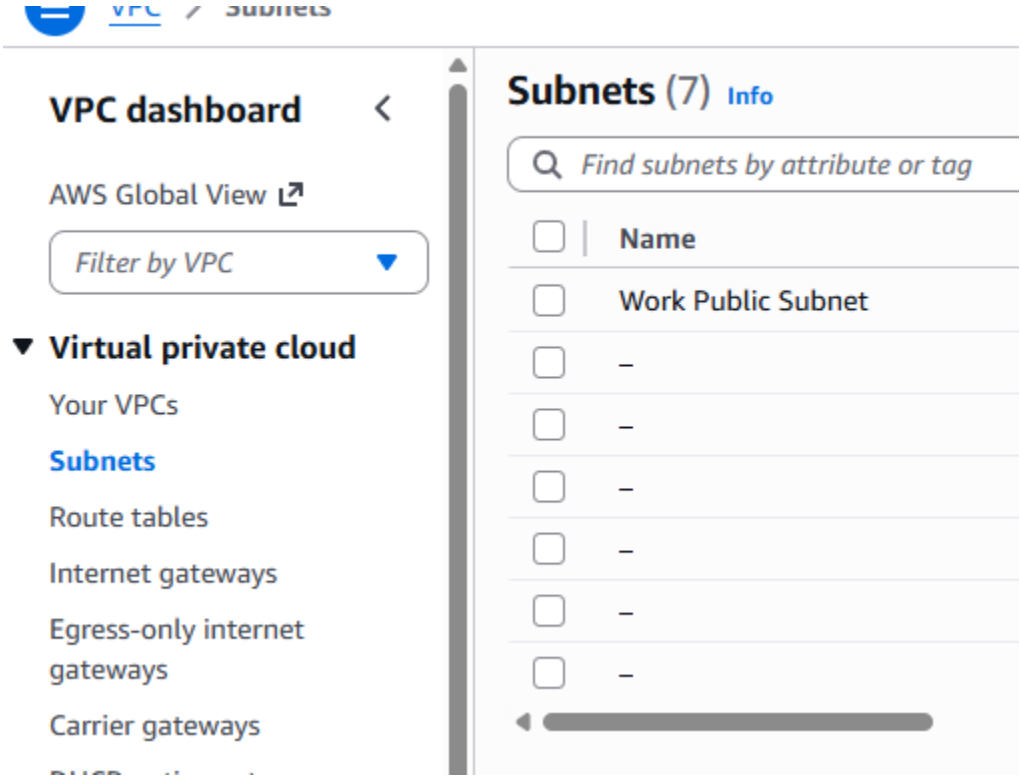
To view the settings of these resources, browse through the VPC console links that display the resource details. For example, choose Subnets to view the subnet details and choose Route tables to view the route table details. The diagram below summarizes the VPC resources you have just created and how they are configured





Task 2 - Create subnets.

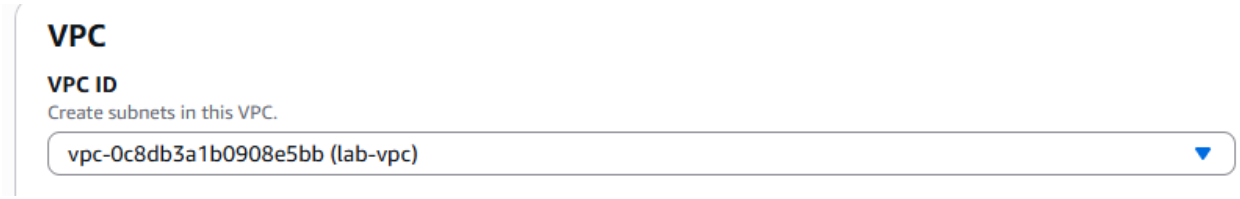
In the left navigation pane, choose Subnets.



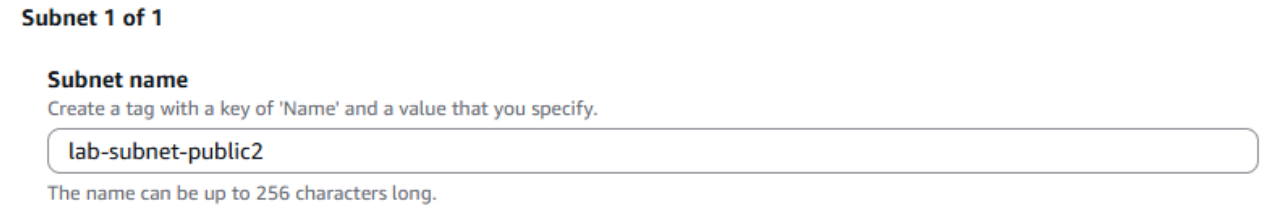
First, you will create a second public subnet

Choose Create subnet then configure:

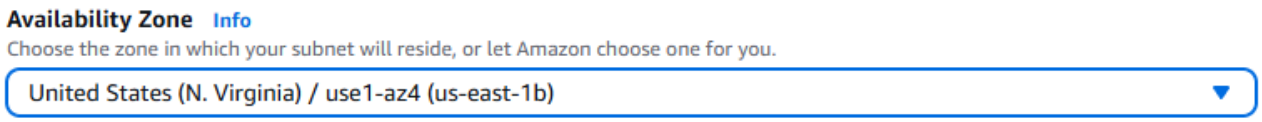
VPC ID: lab-vpc (select from the menu).



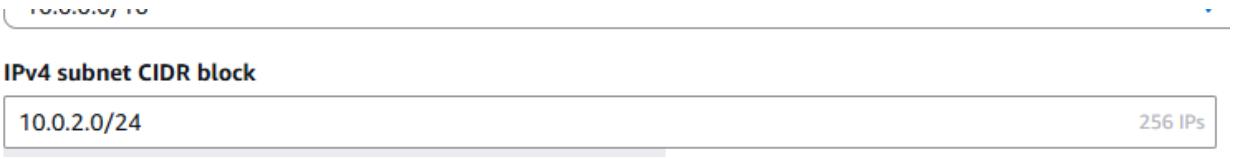
Subnet name: lab-subnet-public2



Availability Zone: Select the second Availability Zone (for example, us-east-1b)

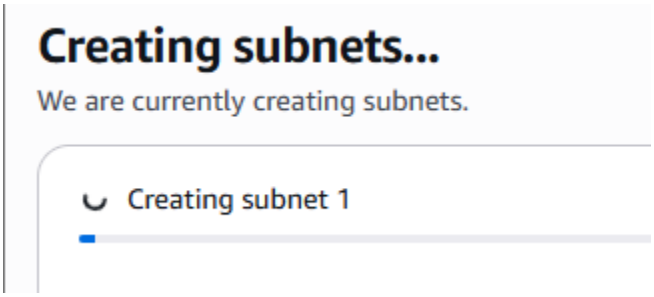


IPv4 CIDR block: 10.0.2.0/24

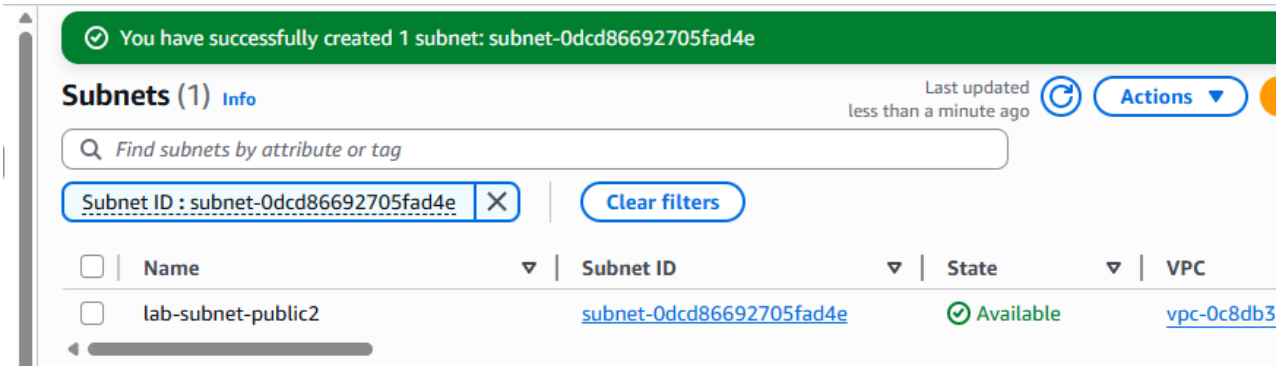


The subnet will have all IP addresses starting with 10.0.2.x.

Choose Create subnet



The second public subnet was created. You will now create a second private subnet.

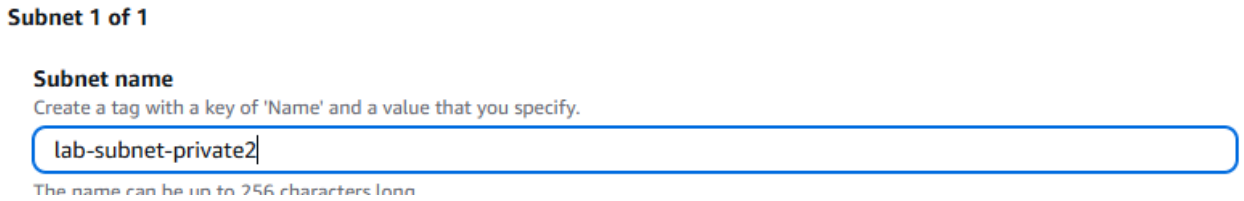


Choose Create subnet then configure:

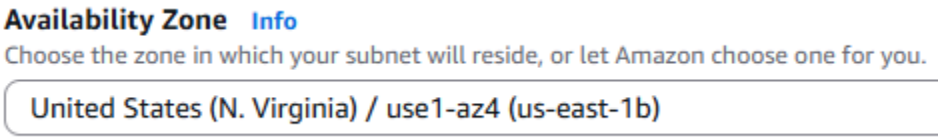
VPC ID: lab-vpc



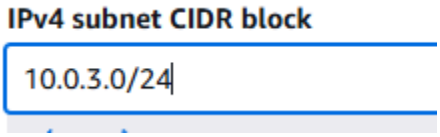
Subnet name: lab-subnet-private2



Availability Zone: Select the second Availability Zone (for example, us-east-1b)

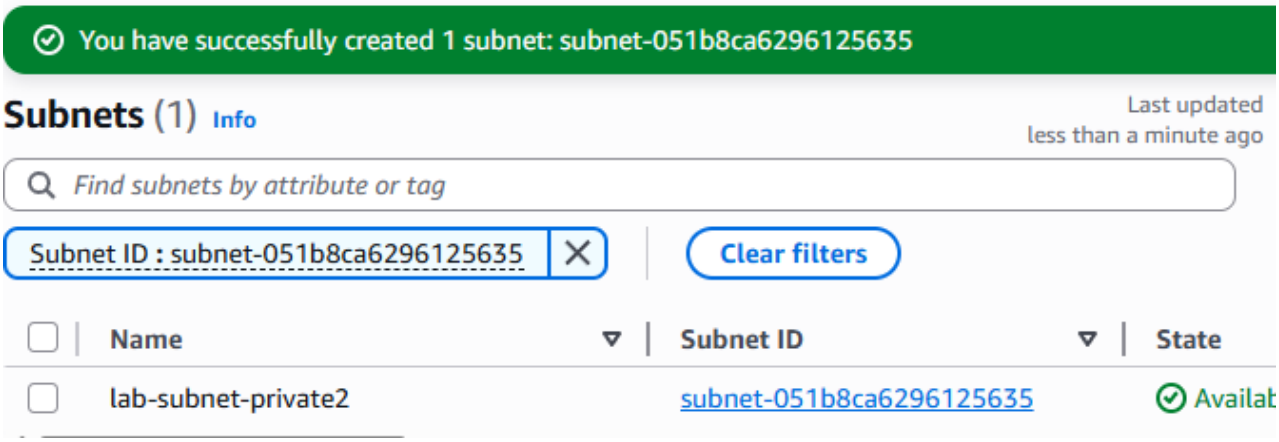


IPv4 CIDR block: 10.0.3.0/24

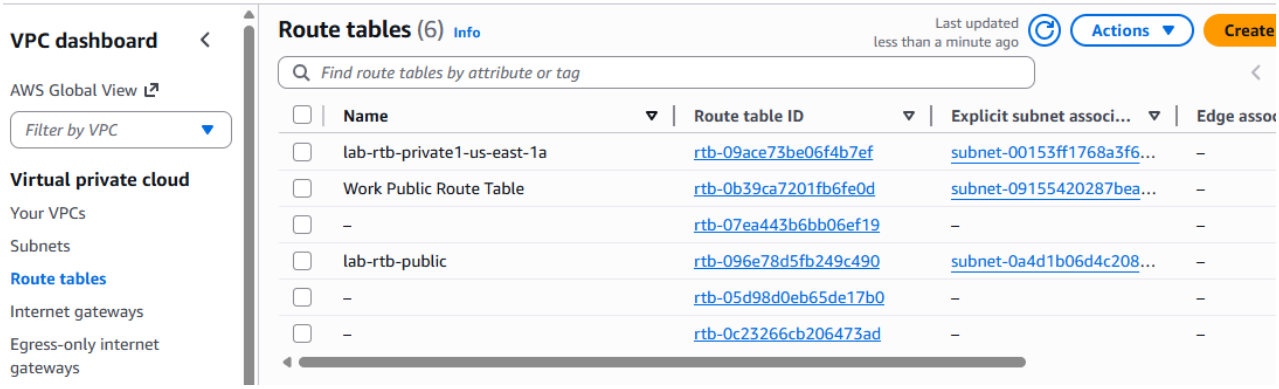


The subnet will have all IP addresses starting with 10.0.3.x

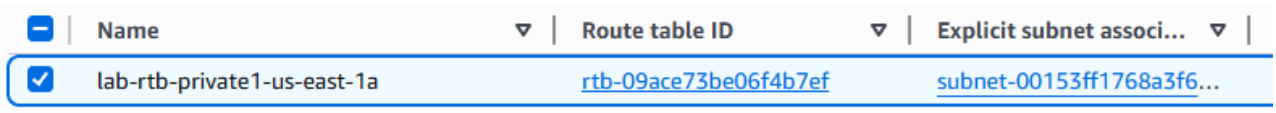
Choose Create subnet



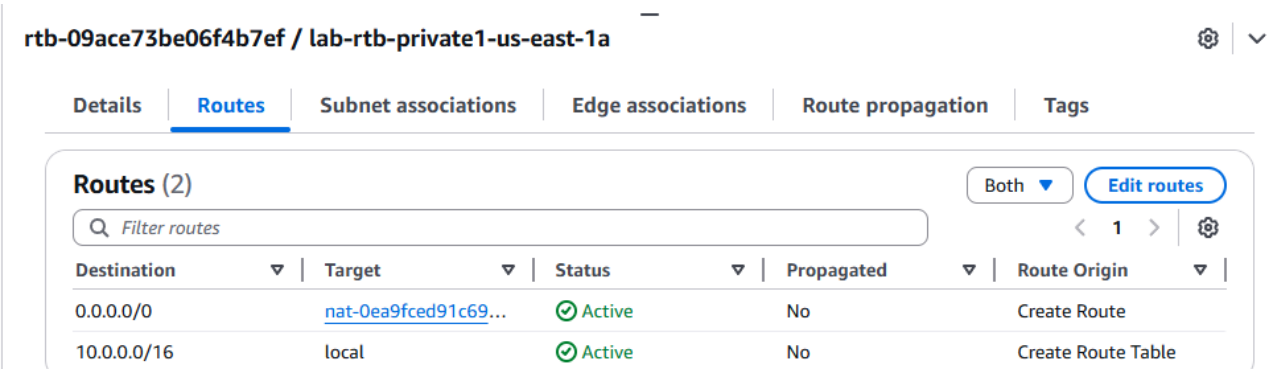
In the left navigation pane, choose **Route tables**



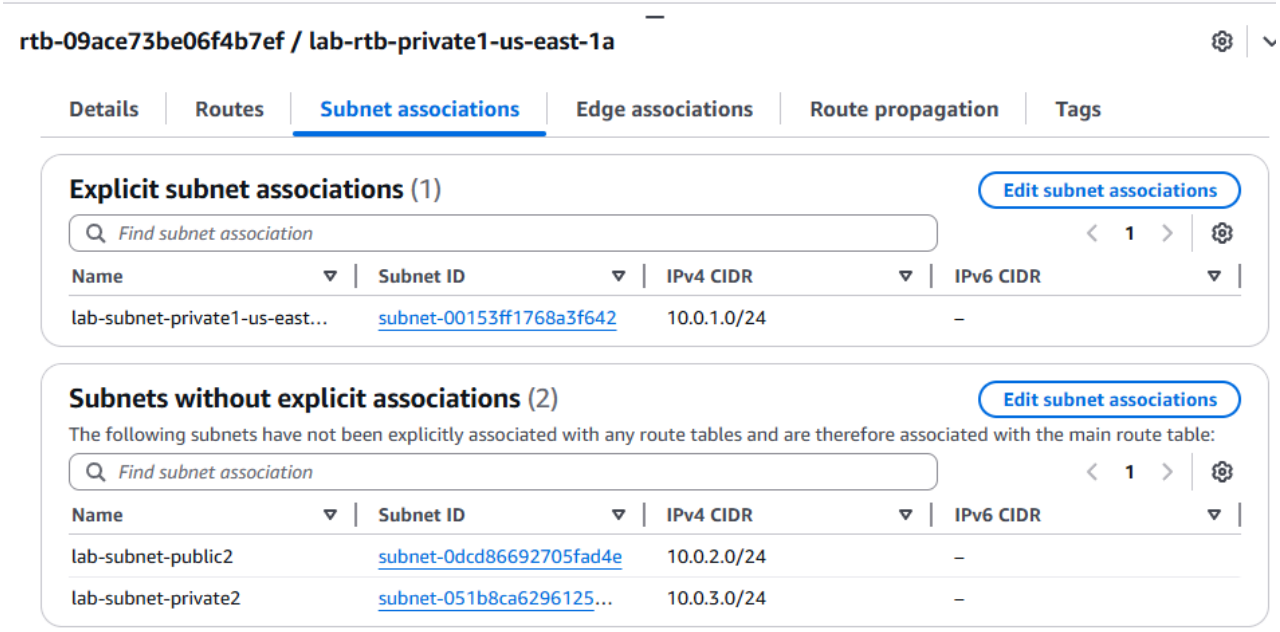
Select the **lab-rtb-private1-us-east-1a** route table



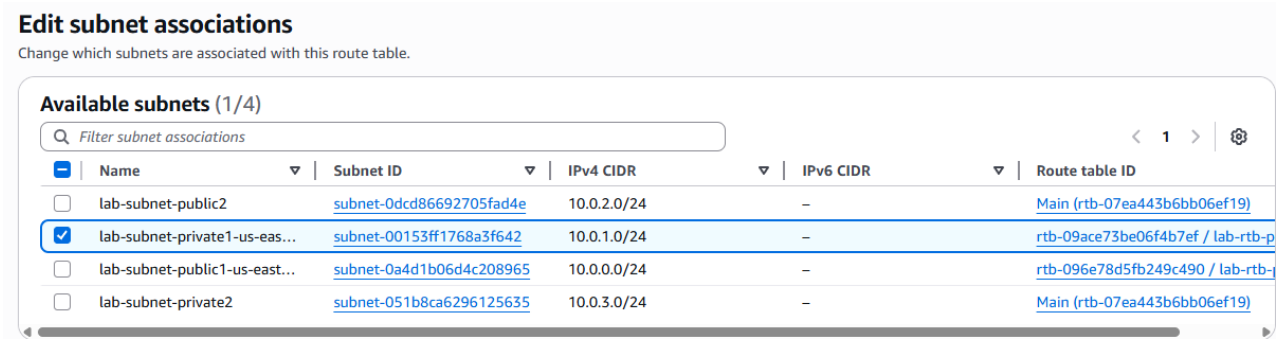
In the lower pane, choose the **Routes** tab



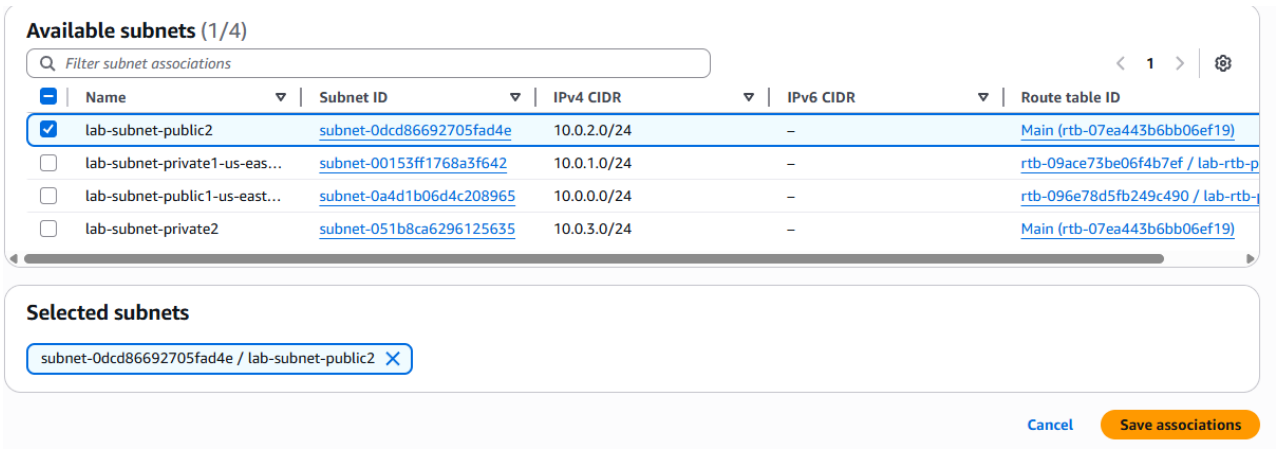
Choose the **Subnet associations** tab



In the Explicit subnet associations panel, choose Edit subnet associations



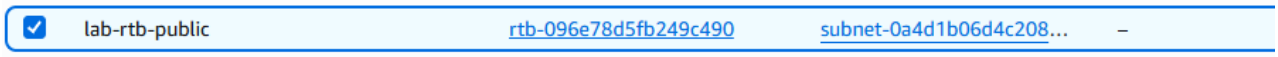
Leave **lab-subnet-private1-us-east-1a** selected, but also select **lab-subnet-private2**



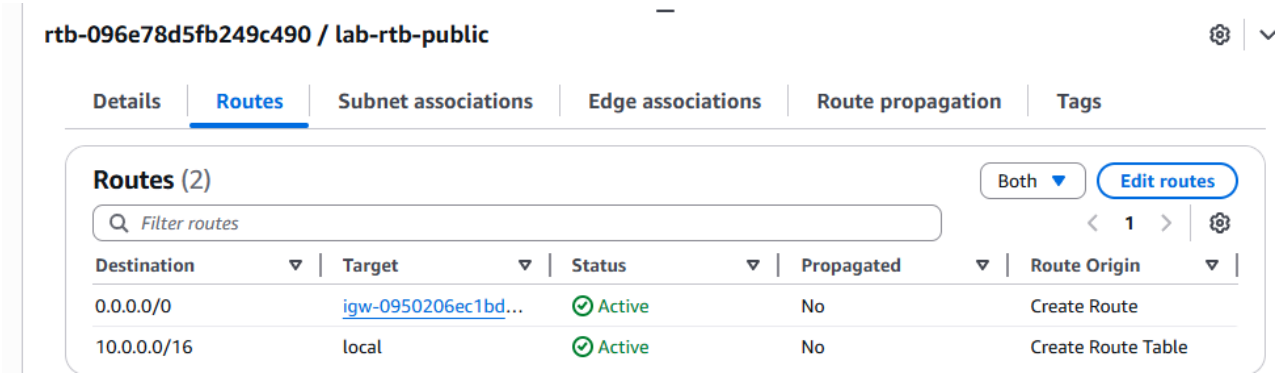
Choose **Save associations**



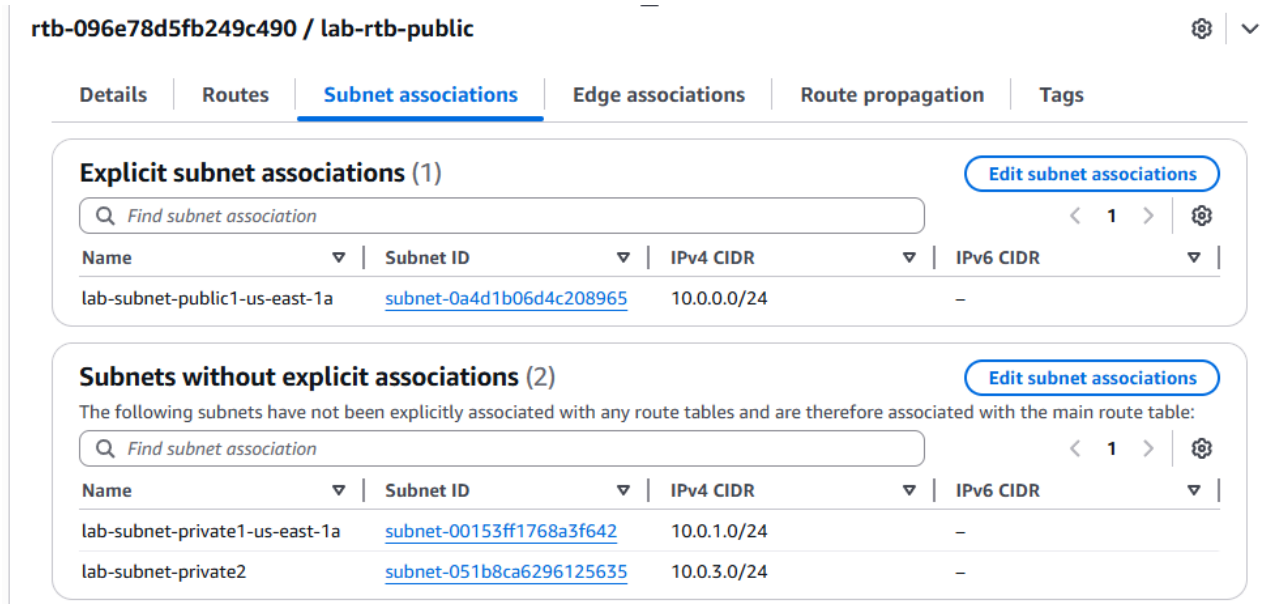
Select the **lab-rtb-public** route table (and deselect any other subnets)



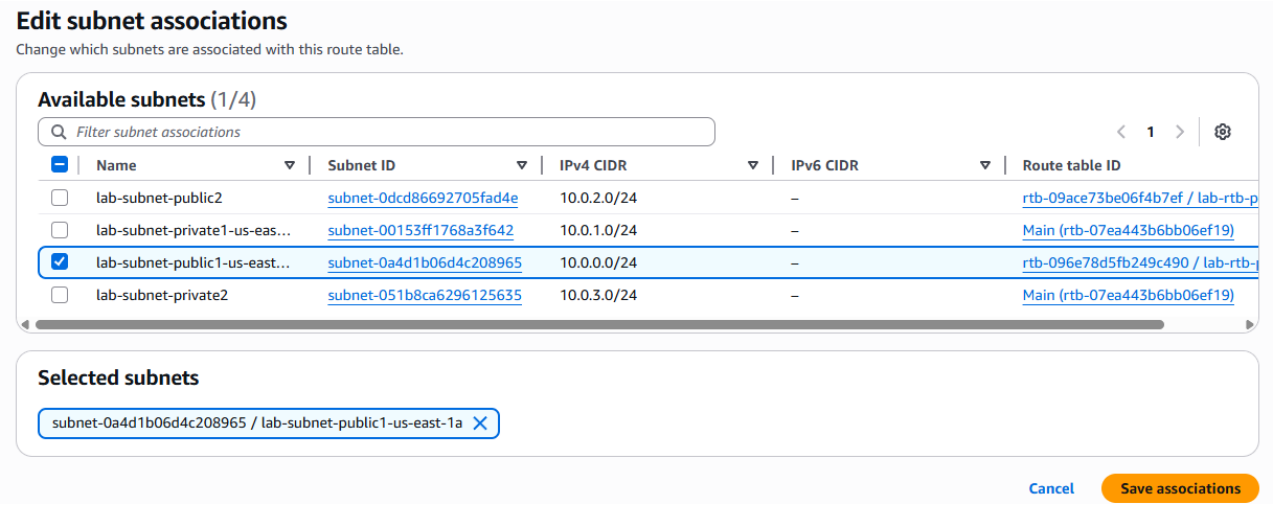
In the lower pane, choose the **Routes** tab



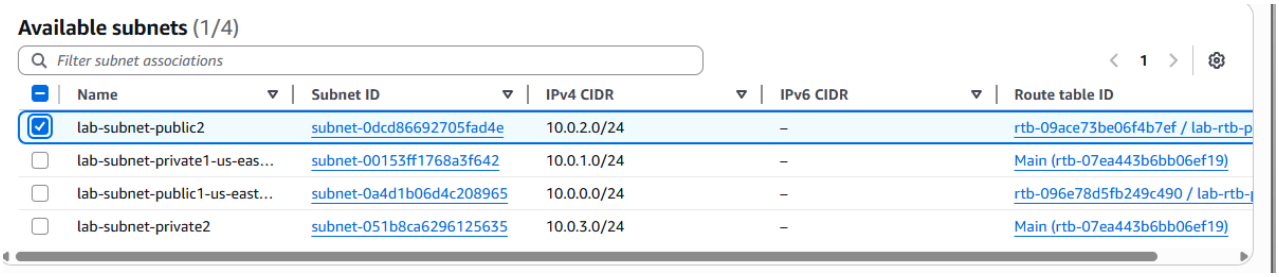
Choose the **Subnet associations** tab



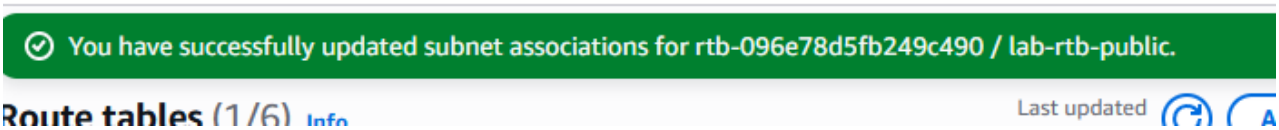
In the Explicit subnet associations area, choose **Edit subnet associations**



Leave **lab-subnet-public1-us-east-1a** selected, but also select **lab-subnet-public2**

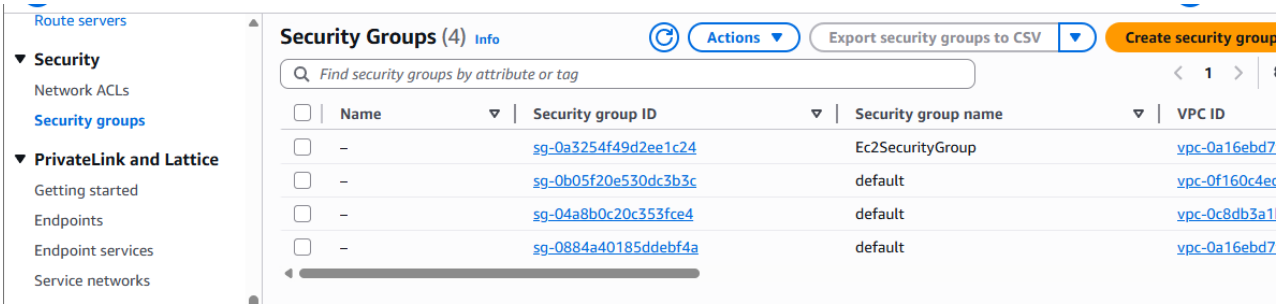


Choose Save associations



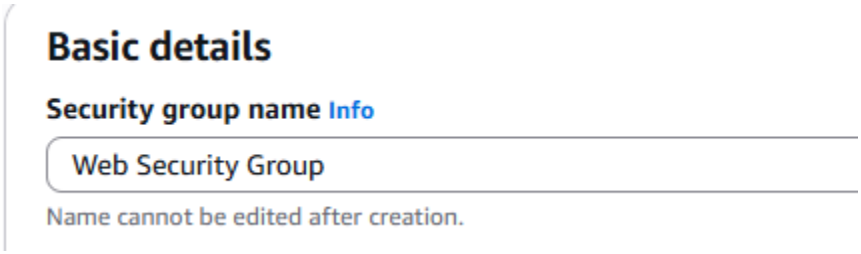
Task 3 - Configure a security group.

In the left navigation pane, choose **Security groups**



Choose Create security group and then configure:

Security group name: Web Security Group



Description: Enable HTTP access



VPC: choose the X to remove the currently selected VPC, then from the drop down list choose lab-vpc



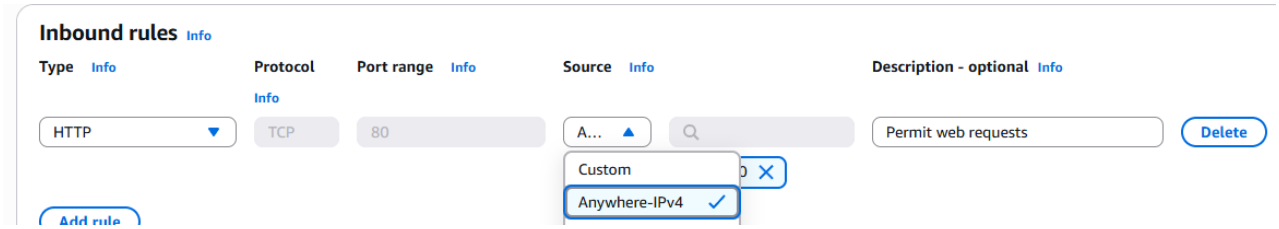
In the Inbound rules pane, choose Add rule

Configure the following settings:

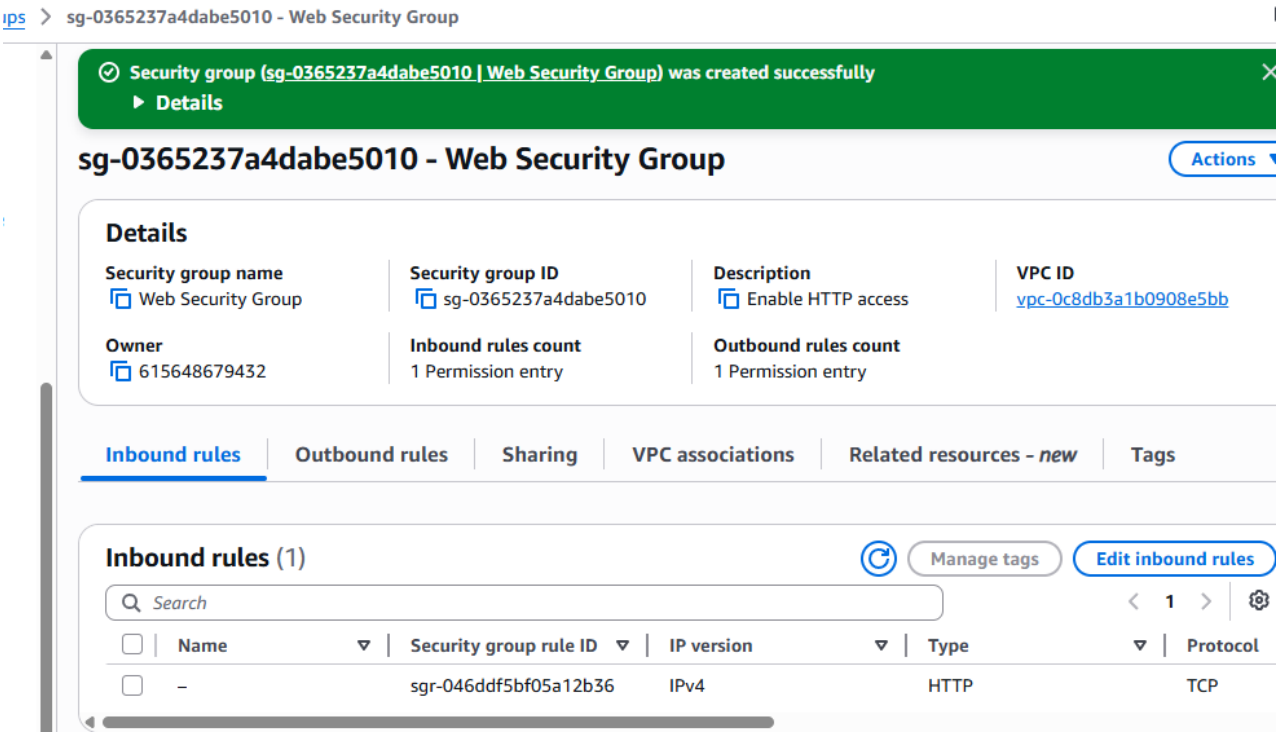
Type: HTTP

Source: Anywhere-IPv4

Description: Permit web requests

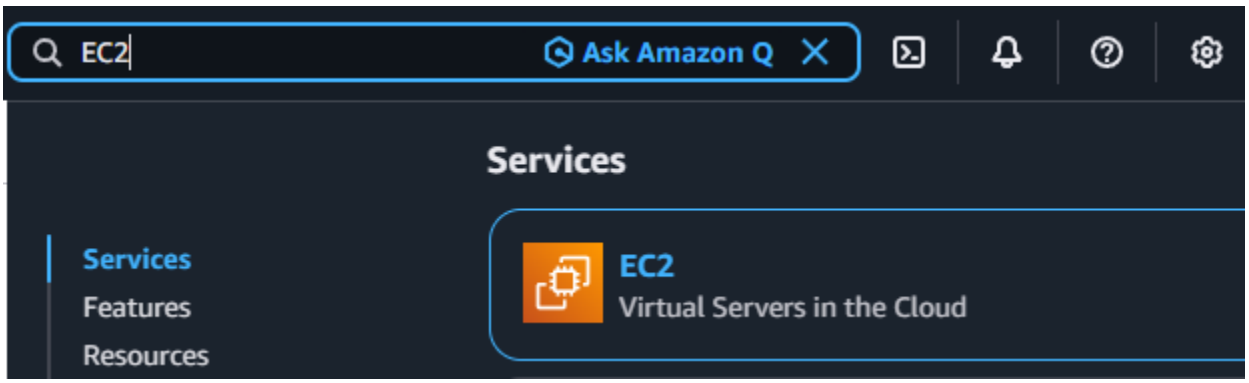


Scroll to the bottom of the page and choose Create security group

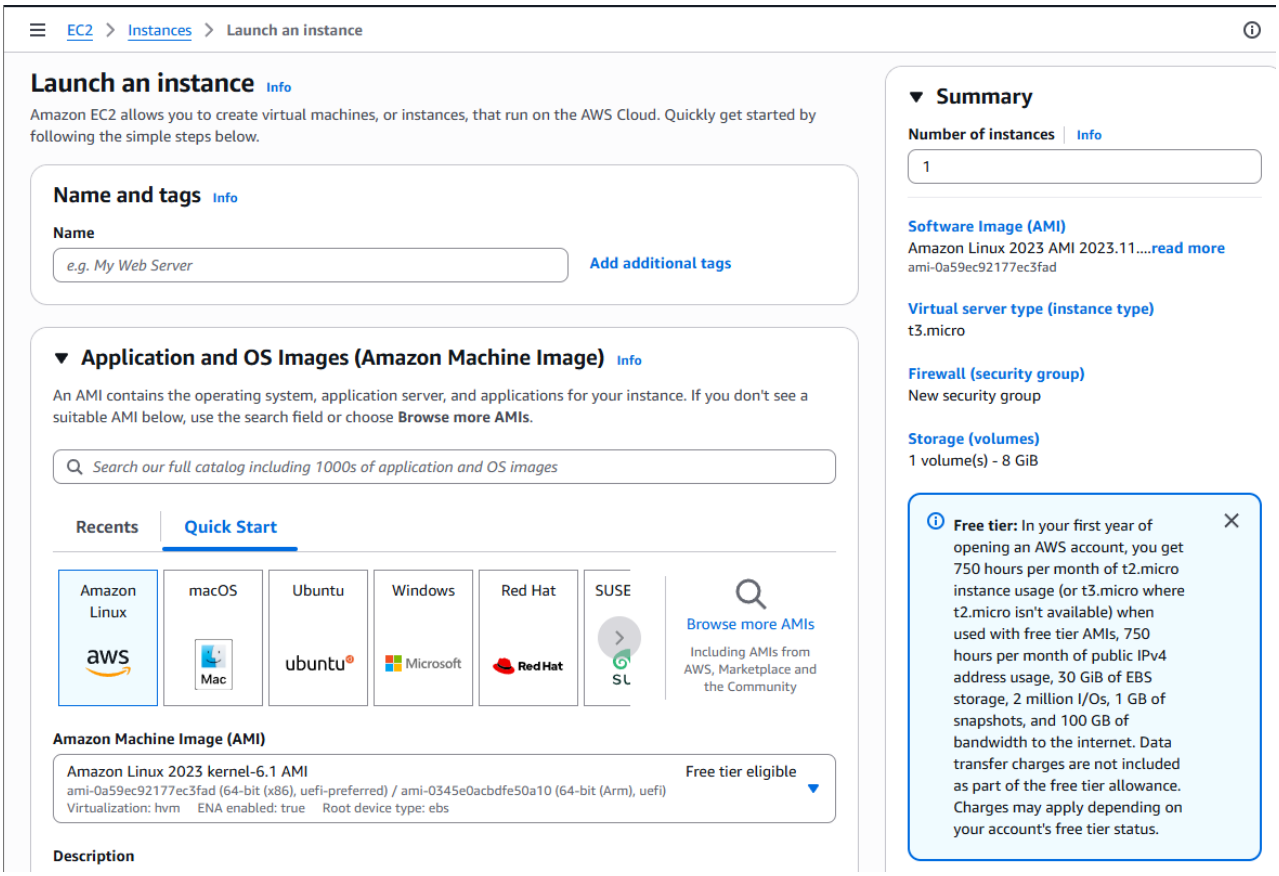


Task 4 - Launch an EC2 instance into a VPC.

In the search box to the right of **Services**, search for and choose **EC2** to open the EC2 console

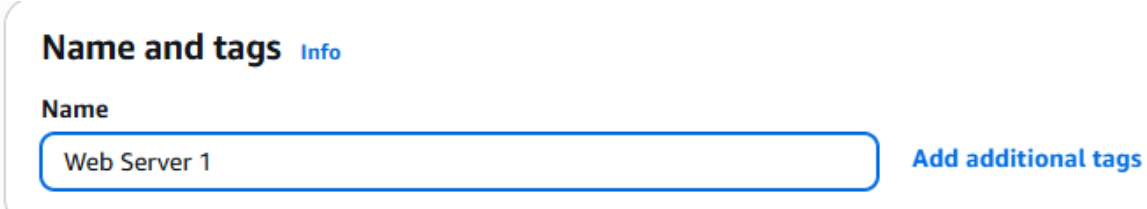


From the Launch instance menu choose Launch instance



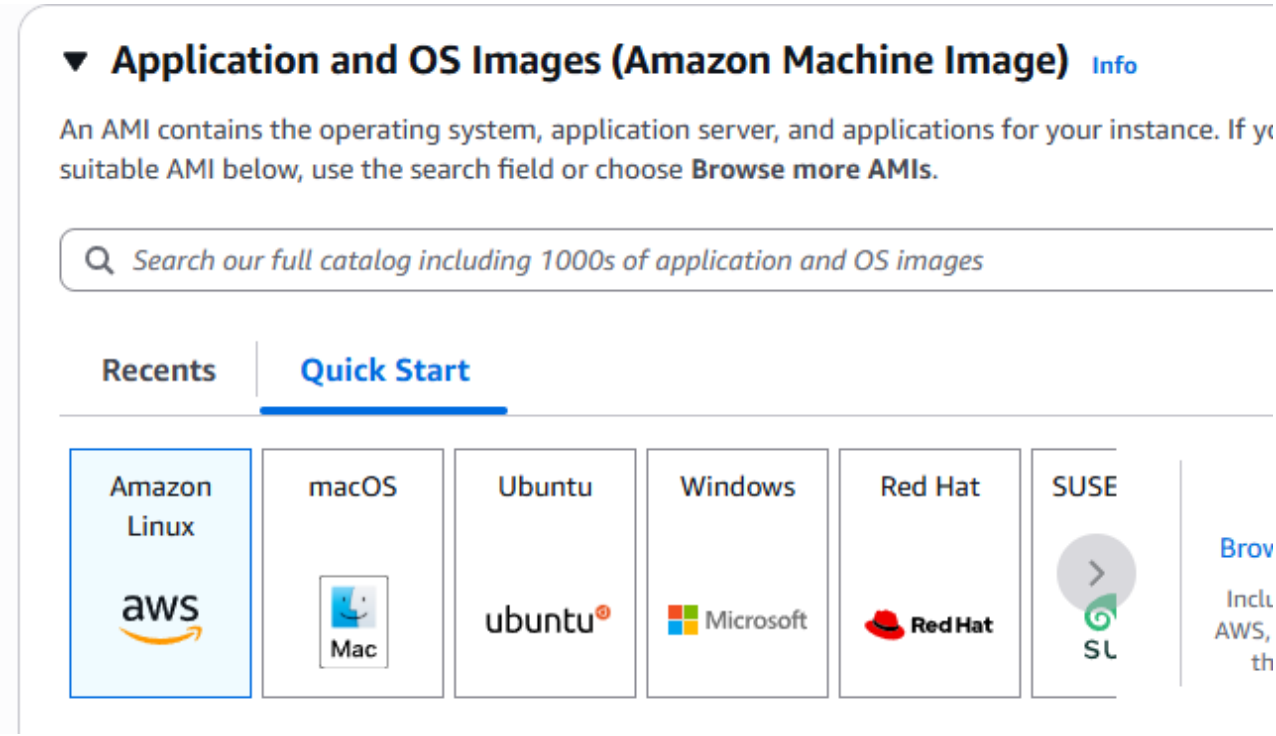
Name the instance:

Give it the name Web Server 1

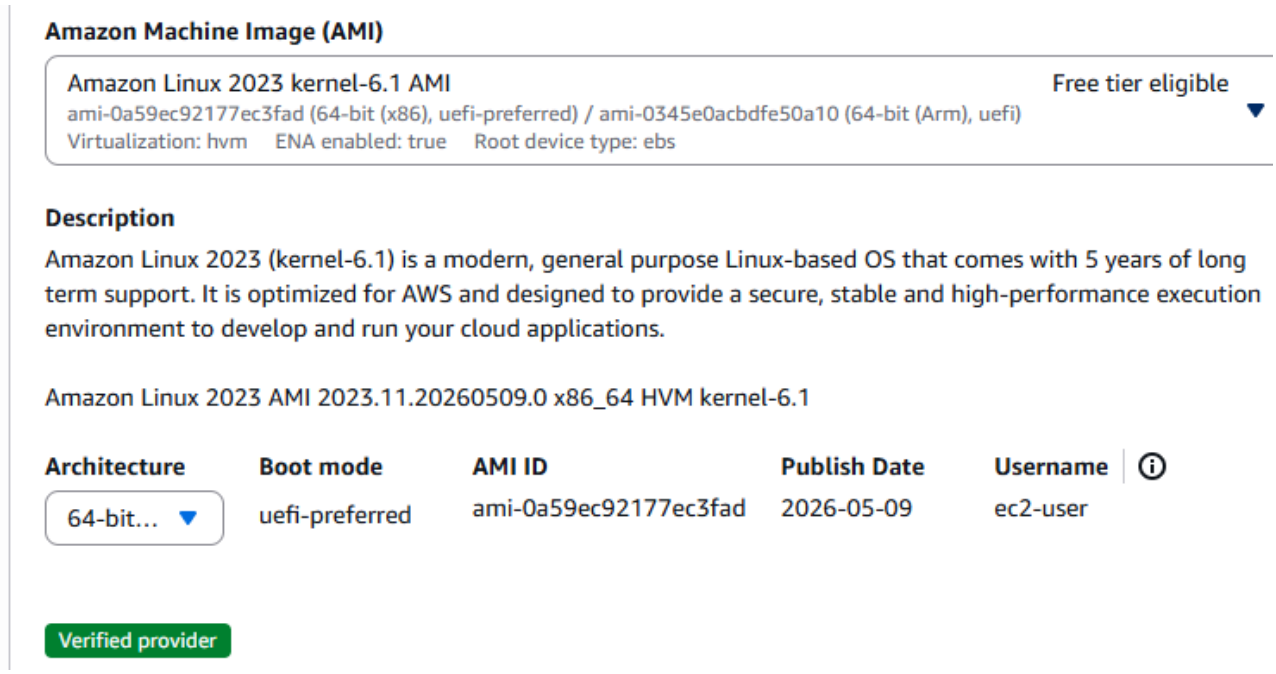


Choose an AMI from which to create the instance:

In the list of available Quick Start AMIs, keep the default Amazon Linux selected.

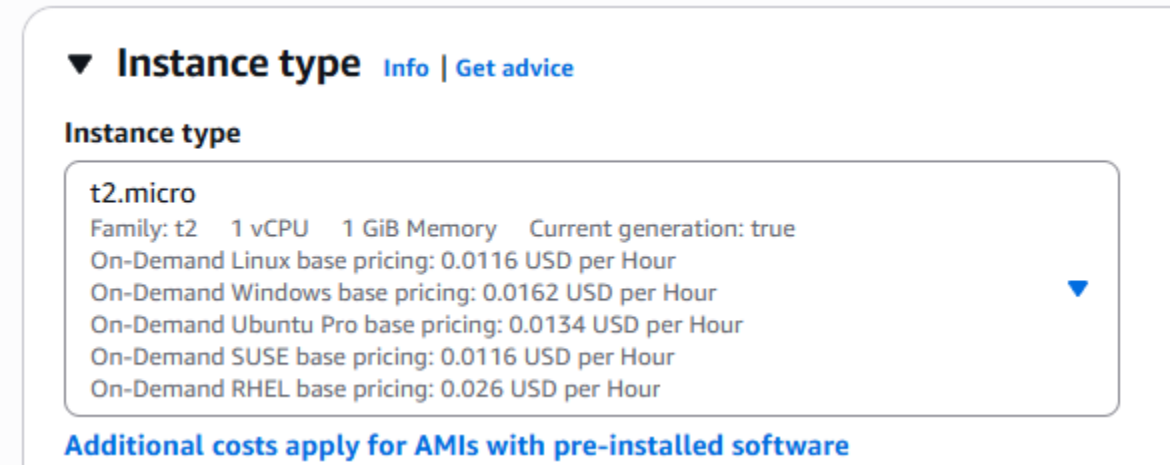


Also keep the default Amazon Linux 2023 AMI selected



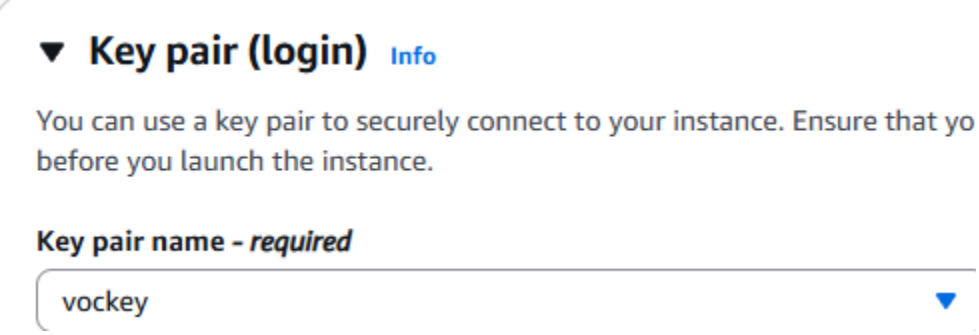
Choose an Instance type:

In the Instance type panel, keep the default t2.micro selected.



Select the key pair to associate with the instance:

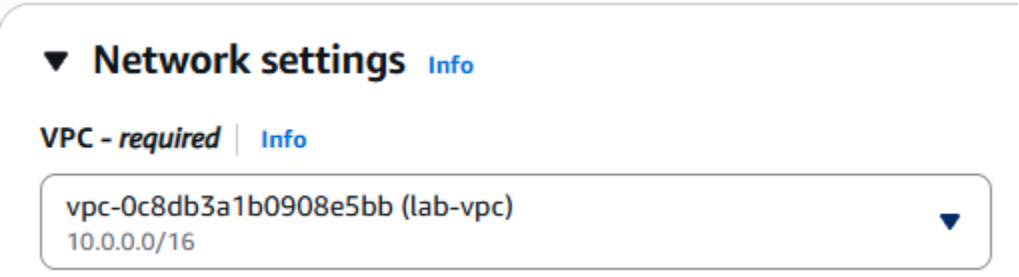
From the Key pair name menu, select vockey



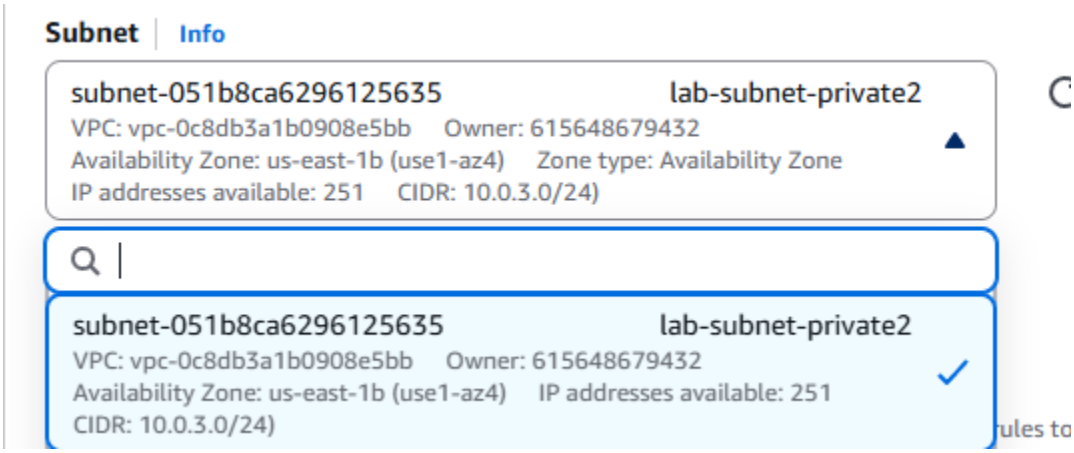
Configure the Network settings:

Next to Network settings, choose Edit, then configure:

Network: lab-vpc



Subnet: lab-subnet-public2 (not Private!)

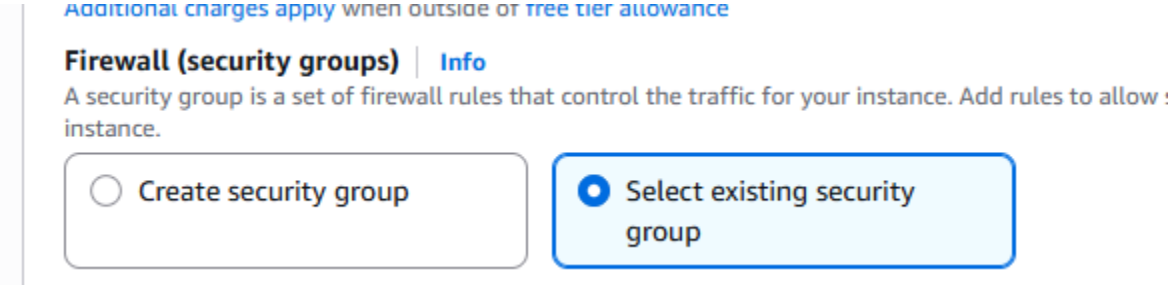


Auto-assign public IP: Enable

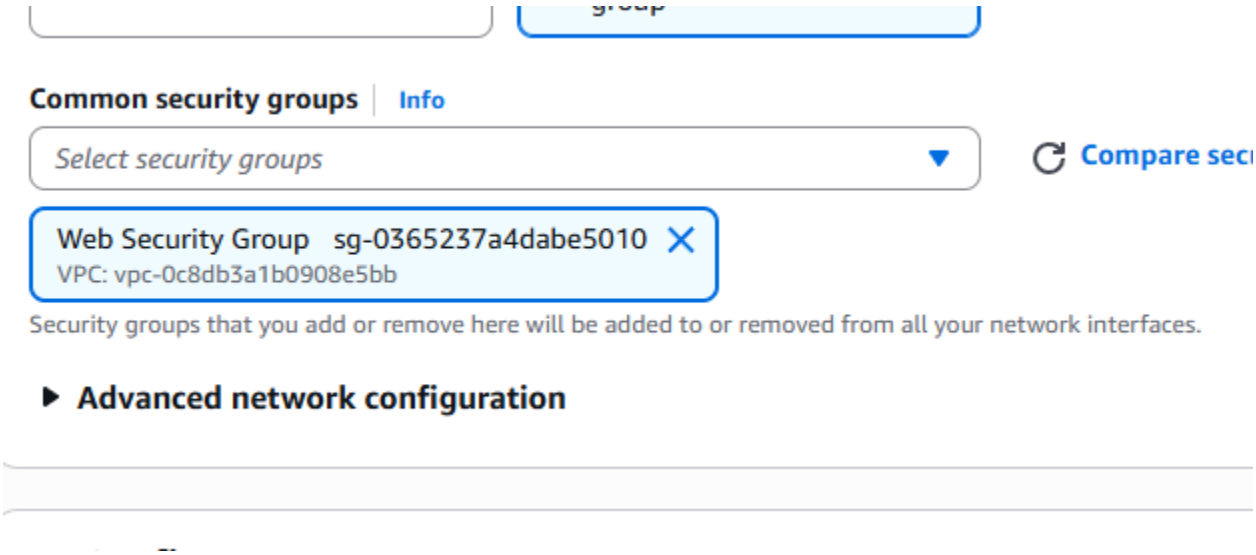


Next, you will configure the instance to use the Web Security Group that you created earlier.

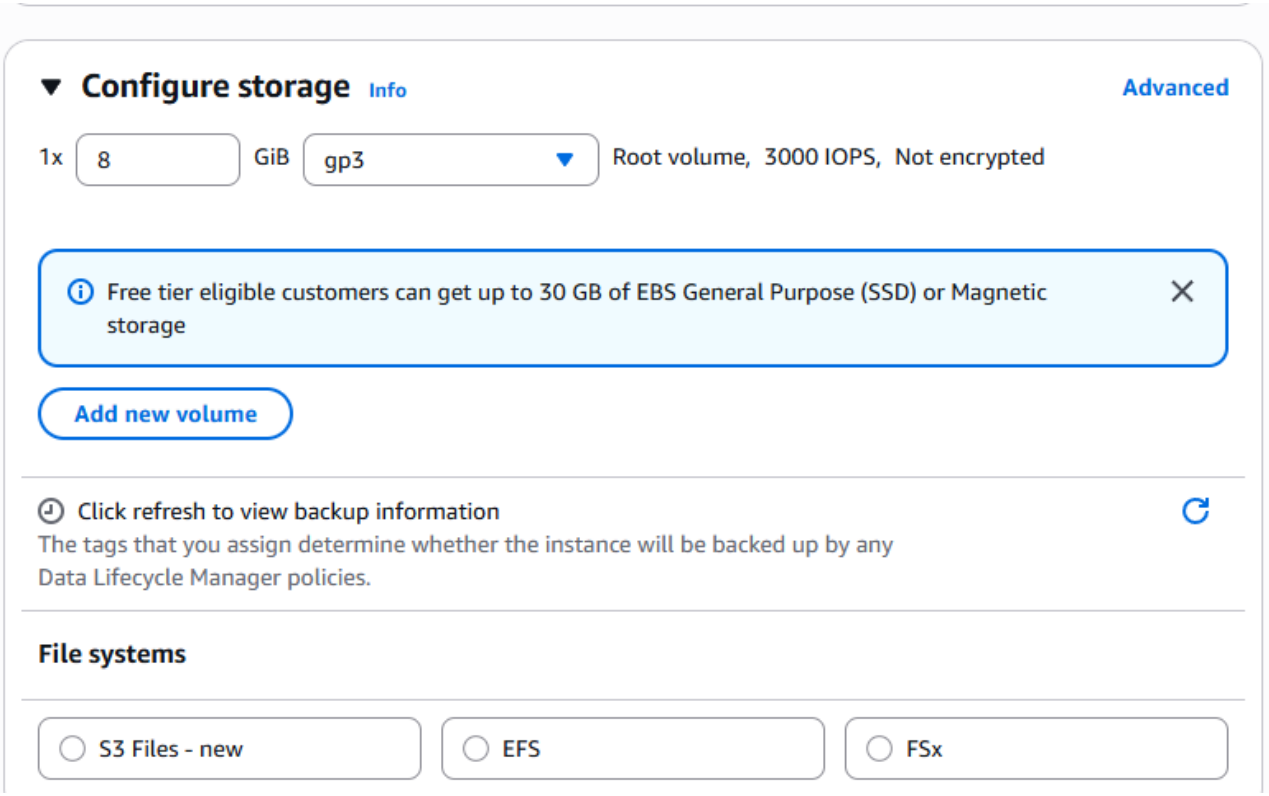
Under Firewall (security groups), choose Select existing security group.



For Common security groups, select Web Security Group



In the *Configure storage* section, keep the default settings



Configure a script to run on the instance when it launches:

Expand the Advanced details panel.

Advanced details Info

Domain join directory Info

Select

Create new directory

IAM instance profile Info

Select

Create new IAM profile

Hostname type Info

IP name

DNS Hostname Info

- [x] Enable IP name IPv4 (A record) DNS requests
- [ ] Enable resource-based IPv4 (A record) DNS requests
- [ ] Enable resource-based IPv6 (AAAA record) DNS requests

Instance auto-recovery Info

Select

Shutdown behavior Info

Stop

Stop - Hibernate behavior Info

Select

Termination protection Info

Select

Stop protection Info

Select

Detailed CloudWatch monitoring Info

Select

Credit specification Info

Standard

Placement group Info

Select

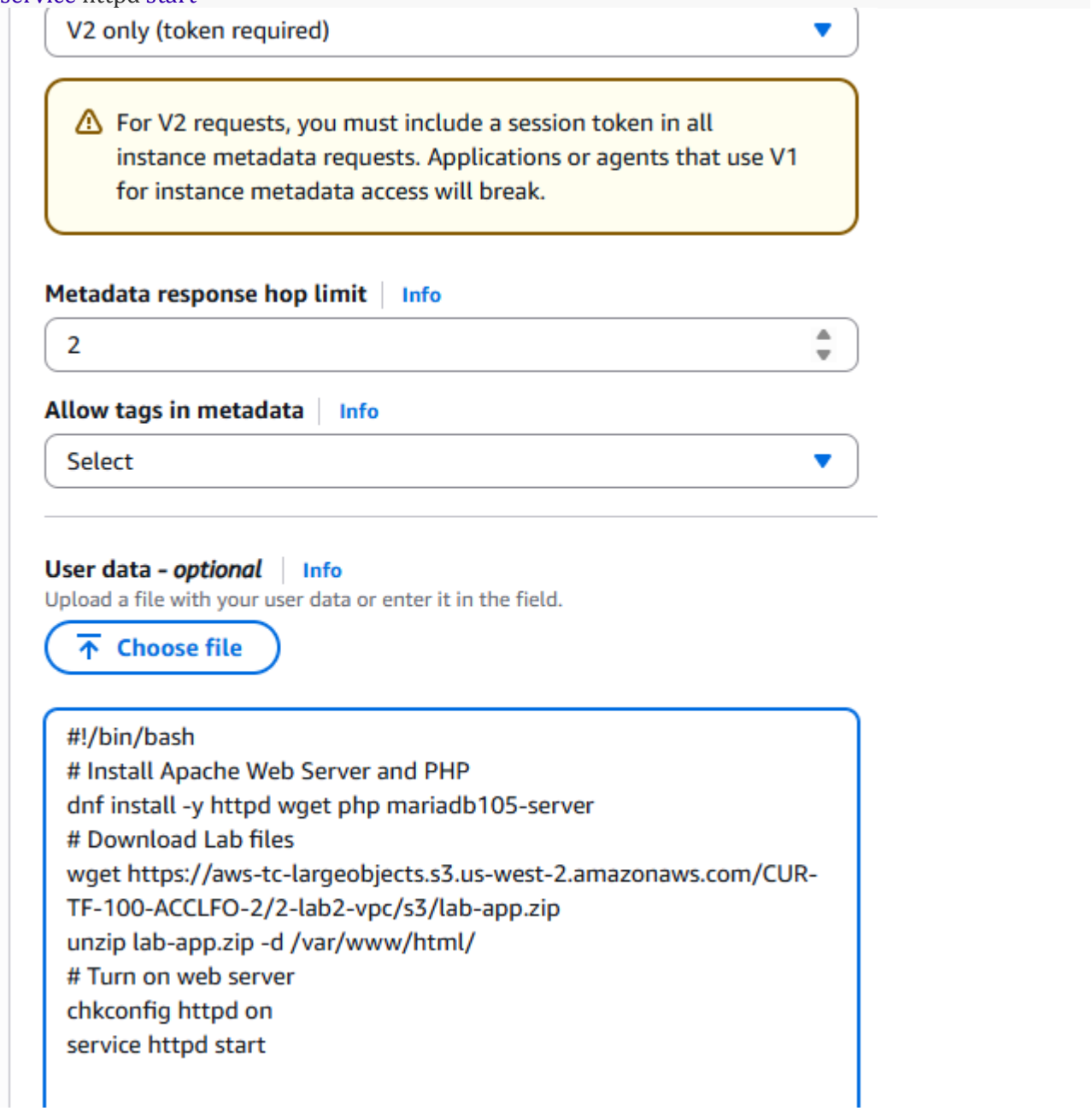
Create new placement group

EBS-optimized instance Info

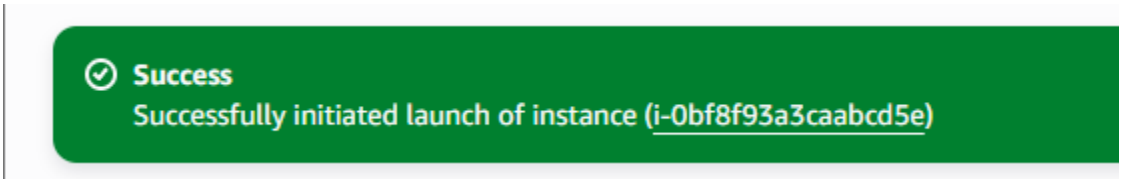
Disable

Scroll to the bottom of the page and then copy and paste the code shown below into the User data box:

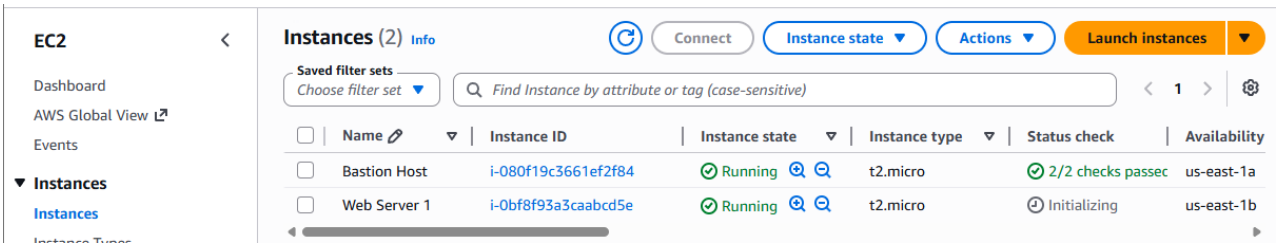
```
#!/bin/bash
# Install Apache Web Server and PHP
dnf install -y httpd wget php mariadb105-server
# Download Lab files
wget https://aws-tc-largeobjects.s3.us-west-2.amazonaws.com/CUR-TF-100-ACCLFO-2/2-lab2-vpc/s3/lab-app.zip
unzip lab-app.zip -d /var/www/html/
# Turn on web server
chkconfig httpd on
service httpd start
```



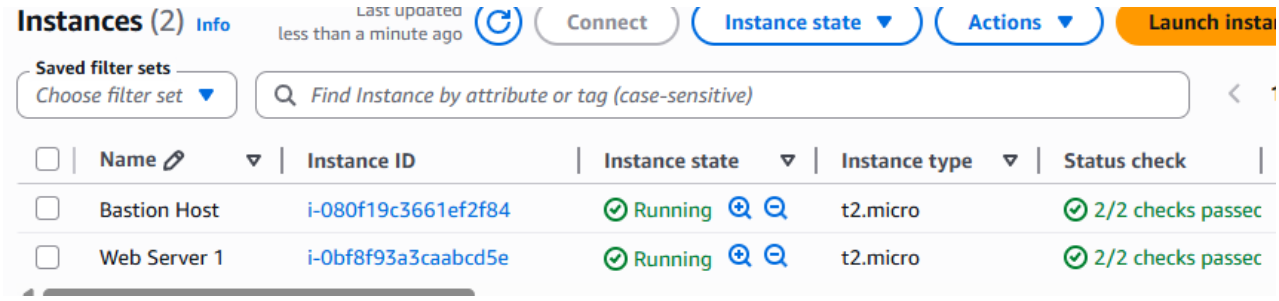
At the bottom of the **Summary** panel on the right side of the screen choose **Launch instance**



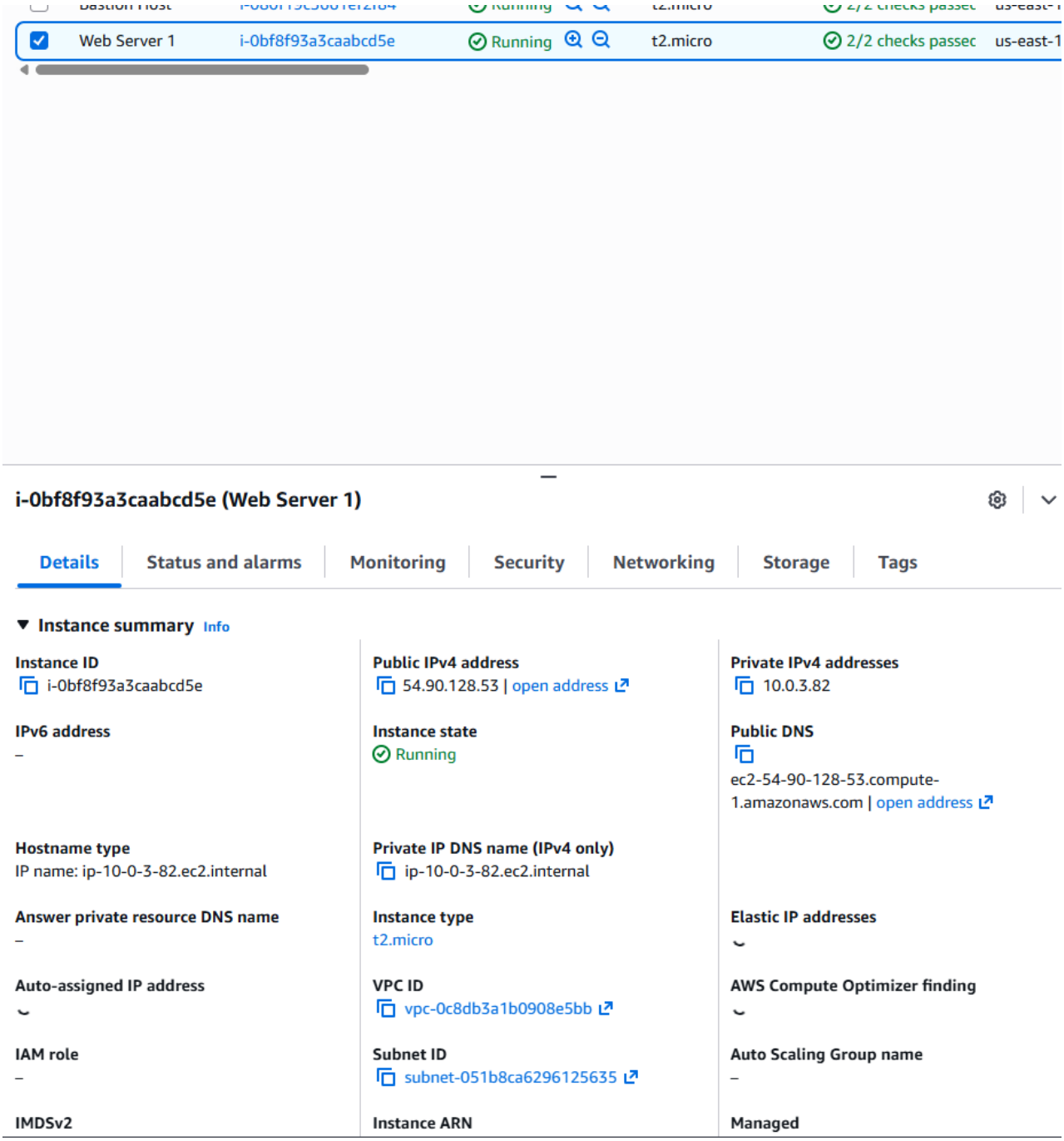
Choose View all instances



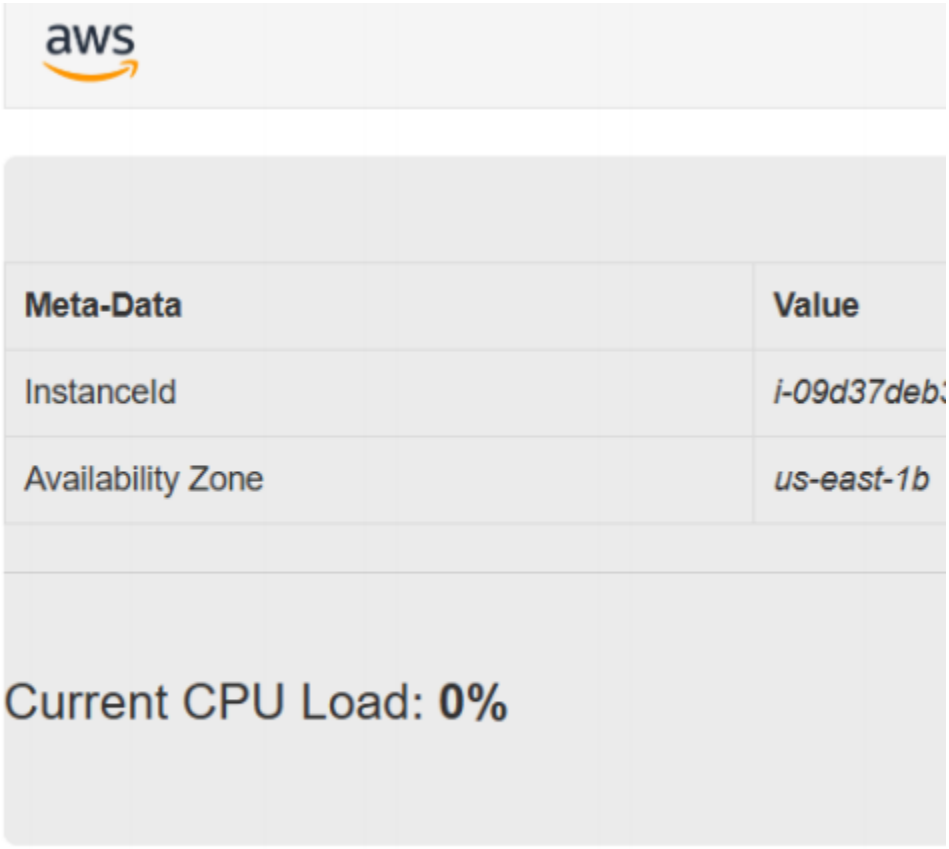
Wait until Web Server 1 shows 2/2 checks passed in the Status check column



Select Web Server 1.



Copy the Public IPv4 DNS value shown in the Details tab at the bottom of the page



Task 5 - Creating a VPC Endpoint for S3 (Follow Lesson - 4 PPT (Create a VPC endpoint for S3 in the us-east1 region instruction)

vpc-s3-endpoint is the name ○ Select AWS services as the service category

○ Select com.amazonaws.us-east-1.s3 as the service name

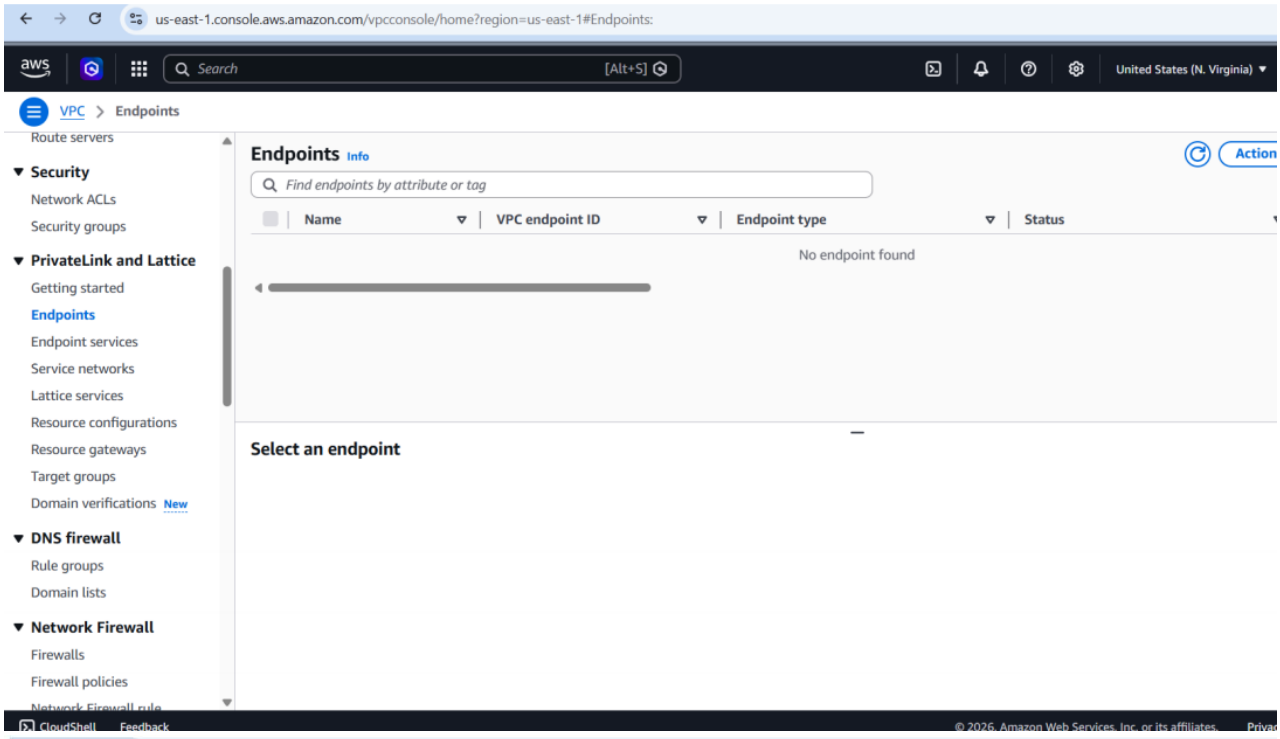
○ Select VPC1 to create the endpoint ○ Select the

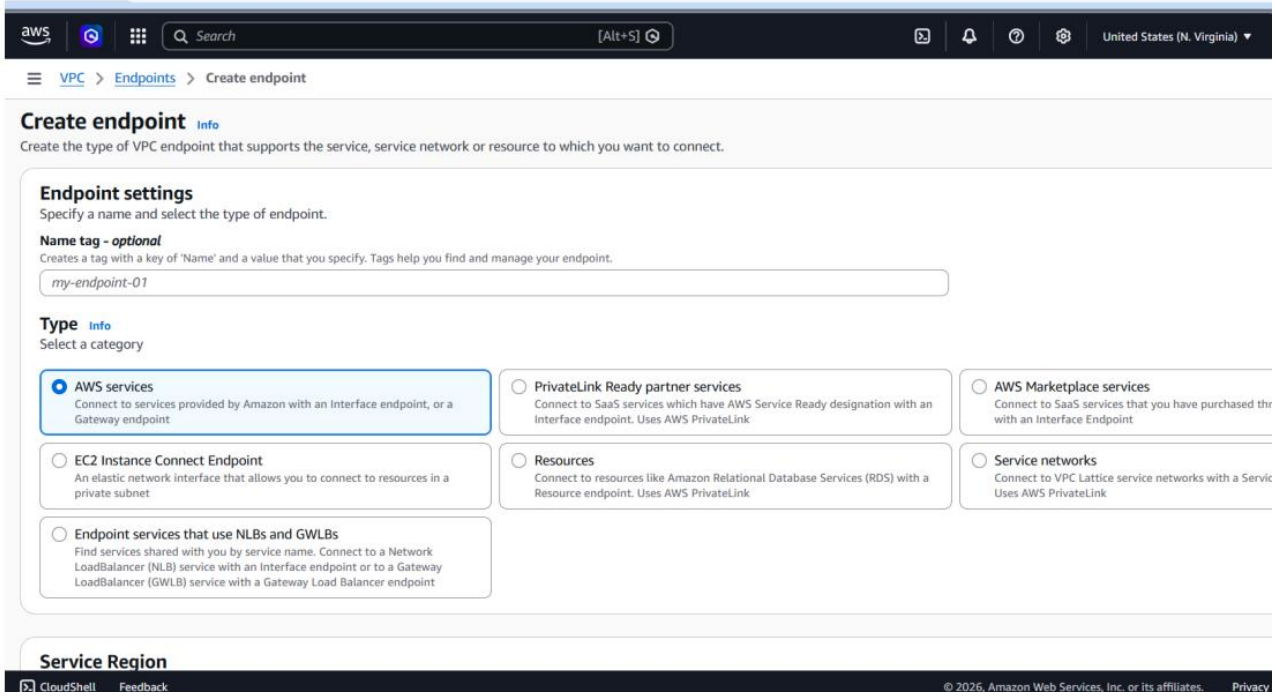
VPC1-Pri route table to associate with the endpoint

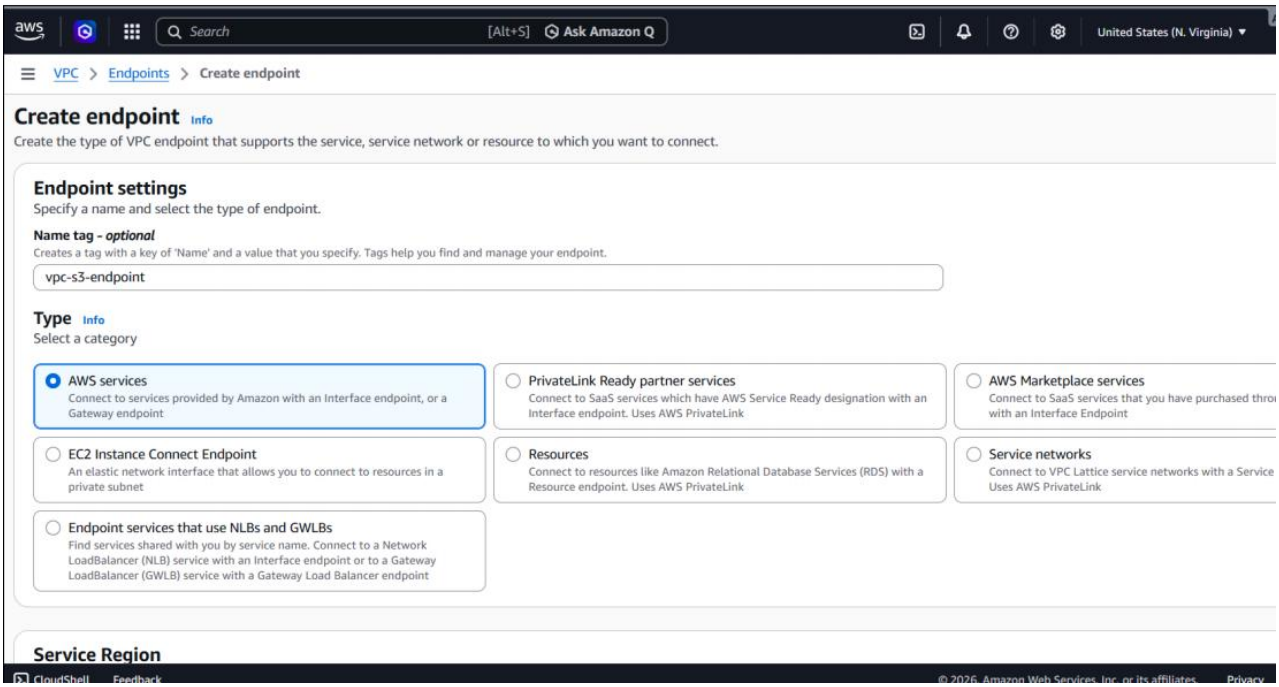
○ Enable policies for resource-based policies to

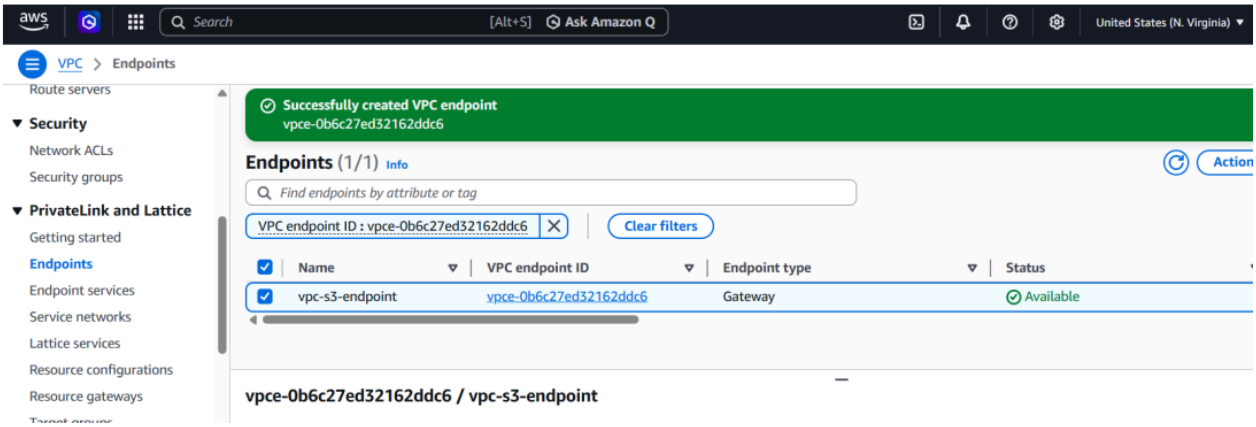
control access to the bucket

○ Click Create endpoint to create the VPC endpoint


us-east-1.console.aws.amazon.com/vpcconsole/home?region=us-east-1#Endpoints:
Search
[Alt+S]
United States (N. Virginia)
VPC > Endpoints
Route servers
Security
Network ACLs
Security groups
PrivateLink and Lattice
Getting started
Endpoints
Endpoint services
Service networks
Lattice services
Resource configurations
Resource gateways
Target groups
Domain verifications New
DNS firewall
Rule groups
Domain lists
Network Firewall
Firewalls
Firewall policies
Endpoints Info
Action
Find endpoints by attribute or tag
Name
VPC endpoint ID
Endpoint type
Status
No endpoint found
Select an endpoint
CloudShell
Feedback
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Search
[Alt+S]
United States (N. Virginia)
VPC > Endpoints > Create endpoint
Create endpoint Info
Create the type of VPC endpoint that supports the service, service network or resource to which you want to connect.
Endpoint settings
Specify a name and select the type of endpoint.
Name tag - optional
Creates a tag with a key of 'Name' and a value that you specify. Tags help you find and manage your endpoint.
my-endpoint-01
Type Info
Select a category
AWS services
Connect to services provided by Amazon with an Interface endpoint, or a Gateway endpoint
PrivateLink Ready partner services
Connect to SaaS services which have AWS Service Ready designation with an Interface endpoint. Uses AWS PrivateLink
AWS Marketplace services
Connect to SaaS services that you have purchased thr
with an Interface Endpoint
EC2 Instance Connect Endpoint
An elastic network interface that allows you to connect to resources in a private subnet
Resources
Connect to resources like Amazon Relational Database Services (RDS) with a Resource endpoint. Uses AWS PrivateLink
Service networks
Connect to VPC Lattice service networks with a Servic
Uses AWS PrivateLink
Endpoint services that use NLBs and GWLBs
Find services shared with you by service name. Connect to a Network LoadBalancer (NLB) service with an Interface endpoint or to a Gateway LoadBalancer (GWLB) service with a Gateway Load Balancer endpoint
Service Region
CloudShell
Feedback
© 2026, Amazon Web Services, Inc. or its affiliates.
Privacy





1. Task 6 – Answering Reflection Questions

- Reflection question responses (Choose any three (3). Must be included in slides with appropriate answer). Provide clear, concise, and well-reasoned answers in your presentation.

1. What is the key difference between an Internet Gateway and a NAT Gateway? Explain the specific purpose each one served in this assignment's architecture.

Answer: An Internet Gateway (IGW) enables two-way communication between resources in public subnets and the public internet external users can initiate inbound connections through it. In this lab, the IGW was attached to the VPC so the public subnet and the NAT Gateway itself could reach the internet.

A NAT Gateway allows instances in private subnets to make outbound requests to the internet without exposing them publicly no external host can initiate a connection inbound through it. In this lab, the NAT Gateway was placed in the public subnet so the private EC2 instance could reach external services (such as S3 or package repositories) without having a public IP

- IGW: bidirectional, used by public subnets, allows inbound traffic from the internet
- NAT Gateway: outbound-only, used by private subnets, blocks unsolicited inbound connections

2. Explain the difference between a Security Group (stateful) and a Network ACL (stateless). Why is it a best practice to use both as a "defense in depth" strategy?

Answer: A Security Group operates at the instance level and is stateful when you allow inbound traffic on a port, the return traffic is automatically permitted without a separate rule. It supports allow rules only. A Network ACL (NACL) operates at the subnet level and is stateless return traffic must be explicitly permitted by its own rule. It supports both allow and deny rules, evaluated in numbered order

Using both together is “defense in depth”: the NACL filters traffic at the subnet boundary before it even reaches an instance. Even if something slips past the NACL, the Security Group provides a second independent check at the instance level. A misconfiguration in one layer does not automatically expose the system because the other layer remains in effect

- Security Group: instance-level, stateful, allow-only rules
- Network ACL: subnet-level, stateless, supports allow and deny rules

3. Why must a NAT Gateway be placed in a public subnet, and what other two components did it depend on to function correctly?

Answer: The NAT Gateway translates private IP addresses into a public Elastic IP so that private instances can reach the internet. For that translation to work, the NAT Gateway itself must have a route to the internet — which only exists in a public subnet (a subnet whose route table sends 0.0.0.0/0 traffic to an Internet Gateway). A private subnet has no such route, so placing the NAT Gateway there would leave it with nowhere to forward the translated packets, breaking outbound access for all private instances

The two components it depended on:

- Internet Gateway (IGW): the public subnet’s route table must point 0.0.0.0/0 to the IGW. Without it, the NAT Gateway can translate addresses but has no actual path to the internet

- Elastic IP Address (EIP): the NAT Gateway must be assigned a static public EIP at creation. This is the address that remote servers see when a private instance makes an outbound request. AWS will not create a NAT Gateway without one.

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Lecturer